

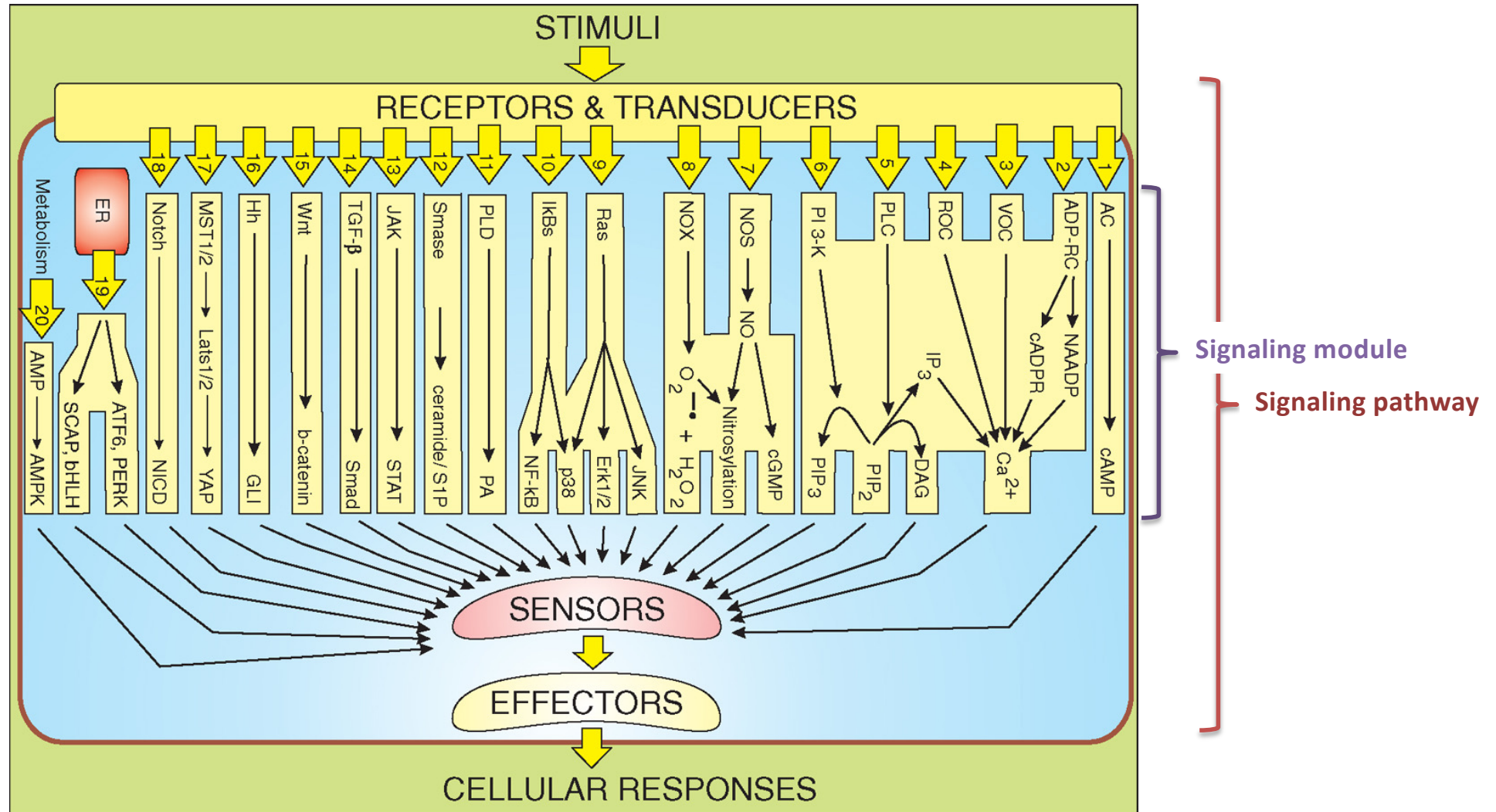
**Molecular Mechanisms of cell signaling (MMCS)
Master Integrative Biology and Physiology
Course for TD #7**

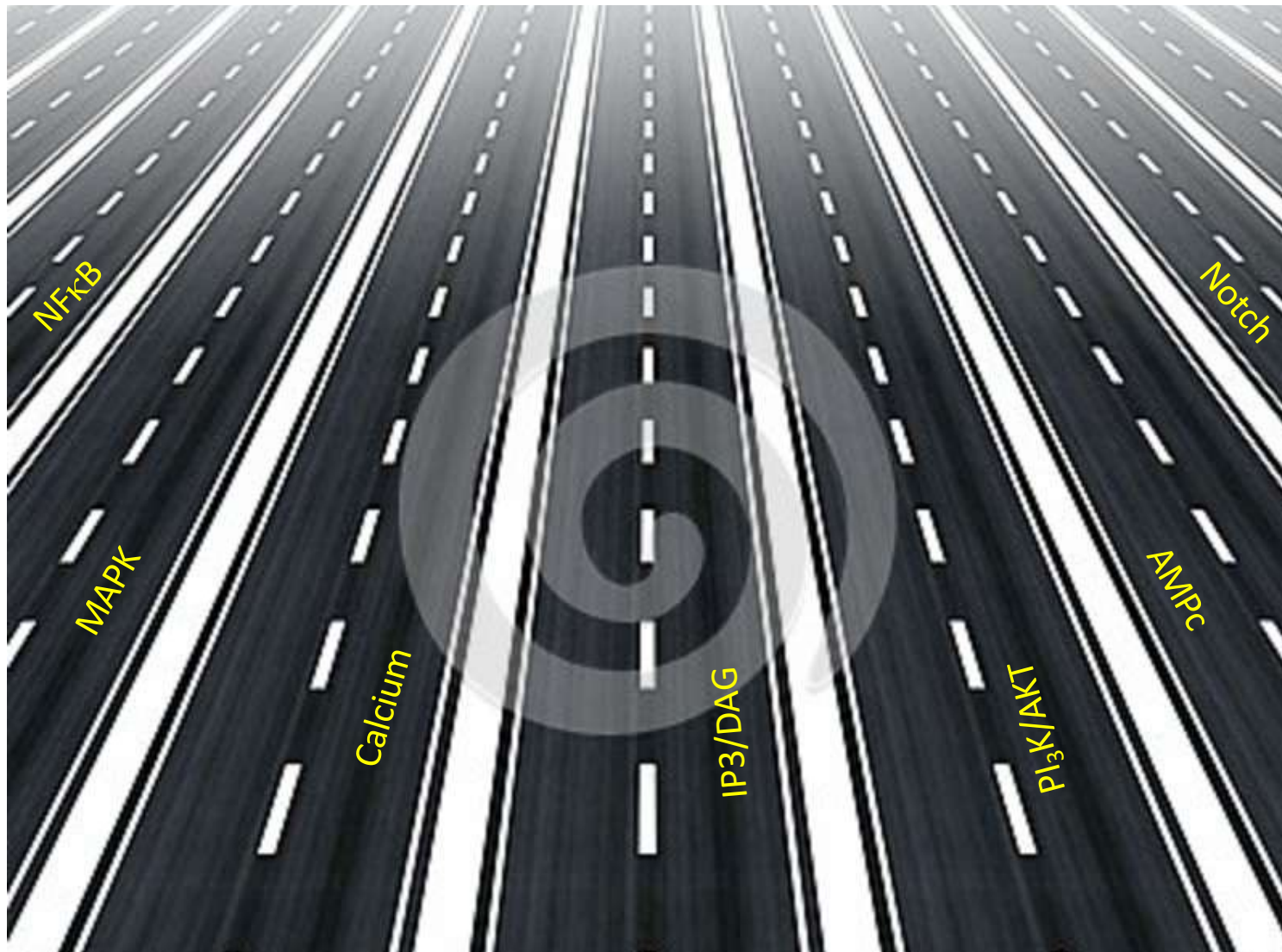
**Signaling integration: Cross-talks of signaling pathways
The example of the NF κ B pathway**

Marc Davenne
marc.davenne@sorbonne-universite.fr

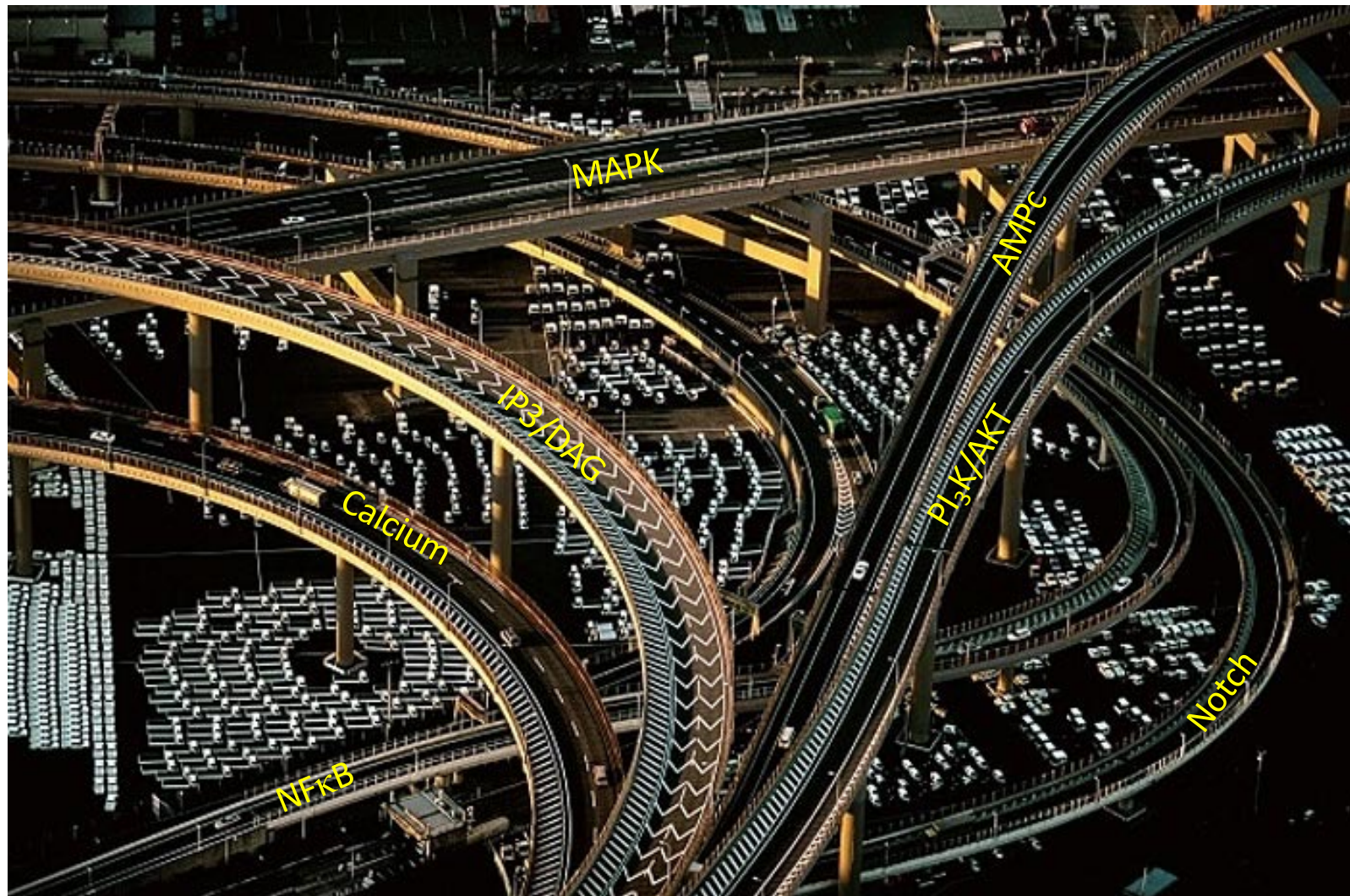
Isabelle Petropoulos
isabelle.petropoulos@sorbonne-universite.fr

What is a signaling pathway ?





Signaling pathways are not quiet long roads leading to one cellular response...

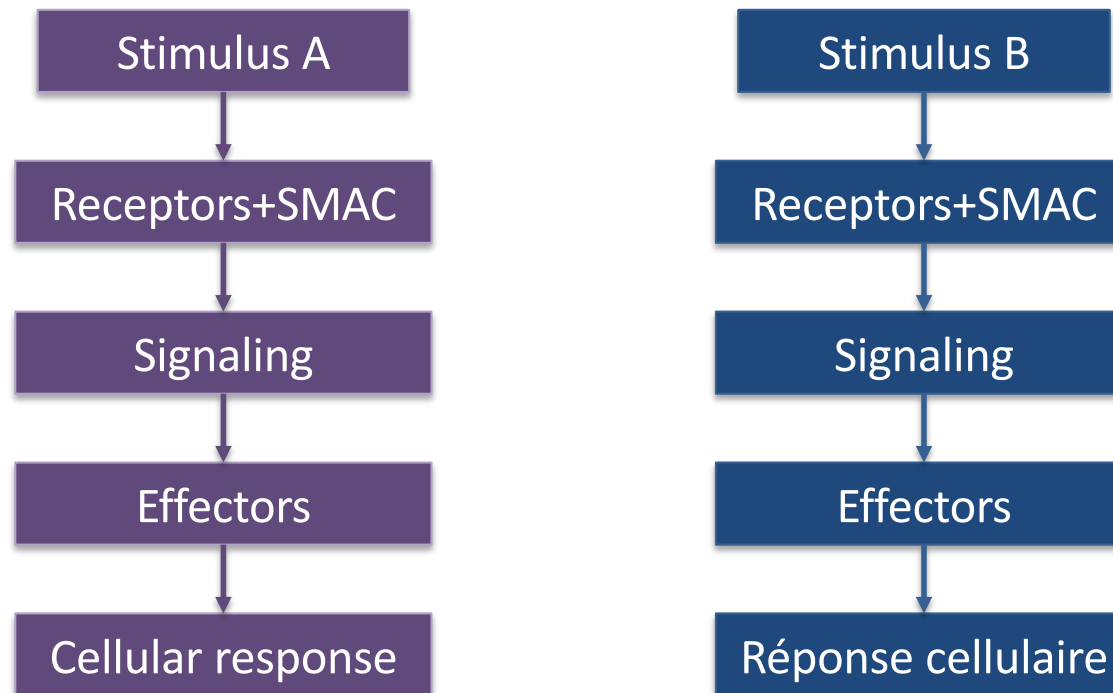


But they interact between each other, and then induce a cellular adaptive response : **Cross-talk**

What is a cross-talk ?

There are only few signaling modules activated regularly and that control a large number of biological processes.

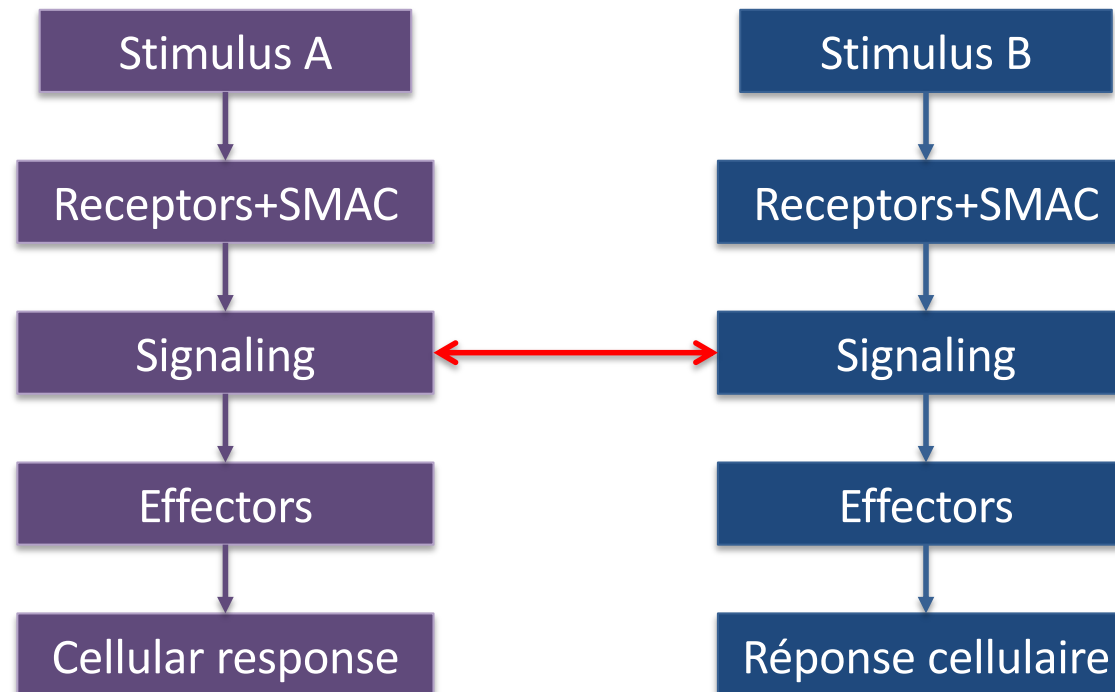
Several members of these signaling pathways are then usually shared by separate pathways, thus resulting in interactions in between pathways.



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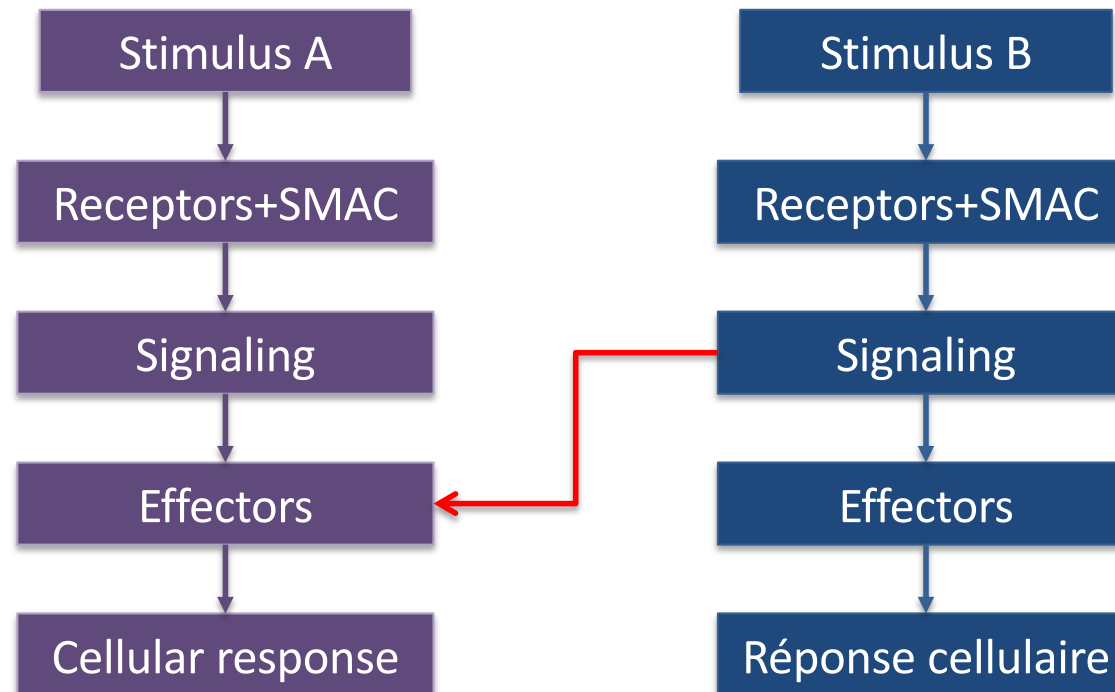
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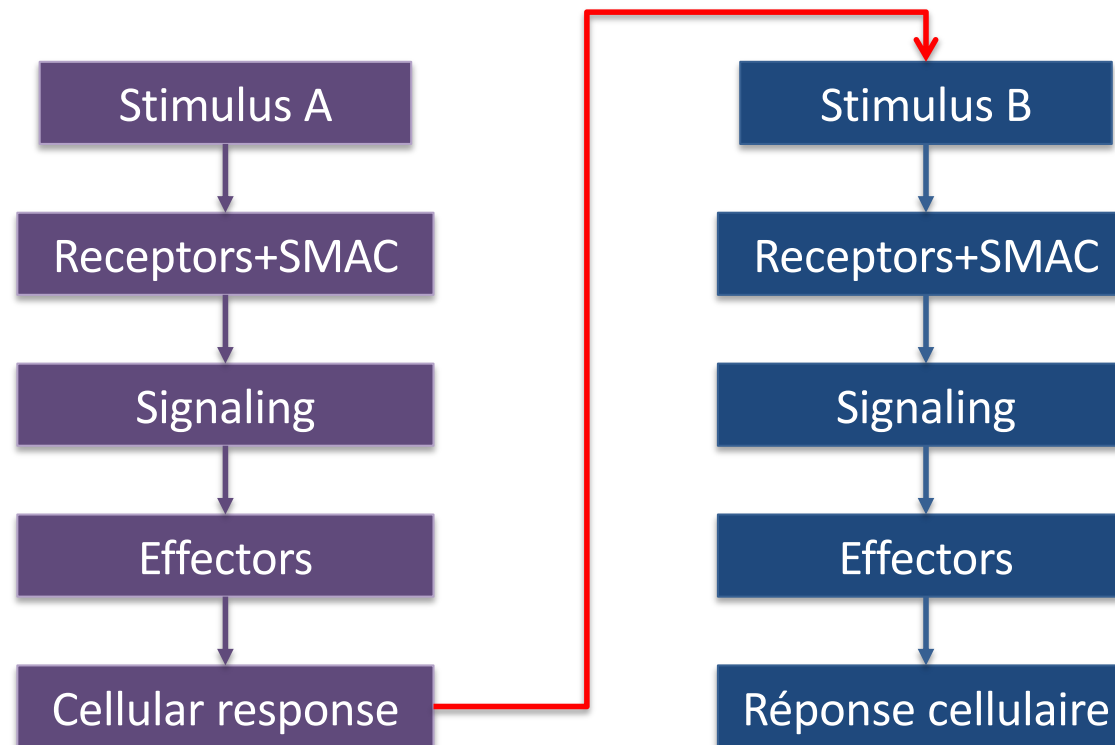
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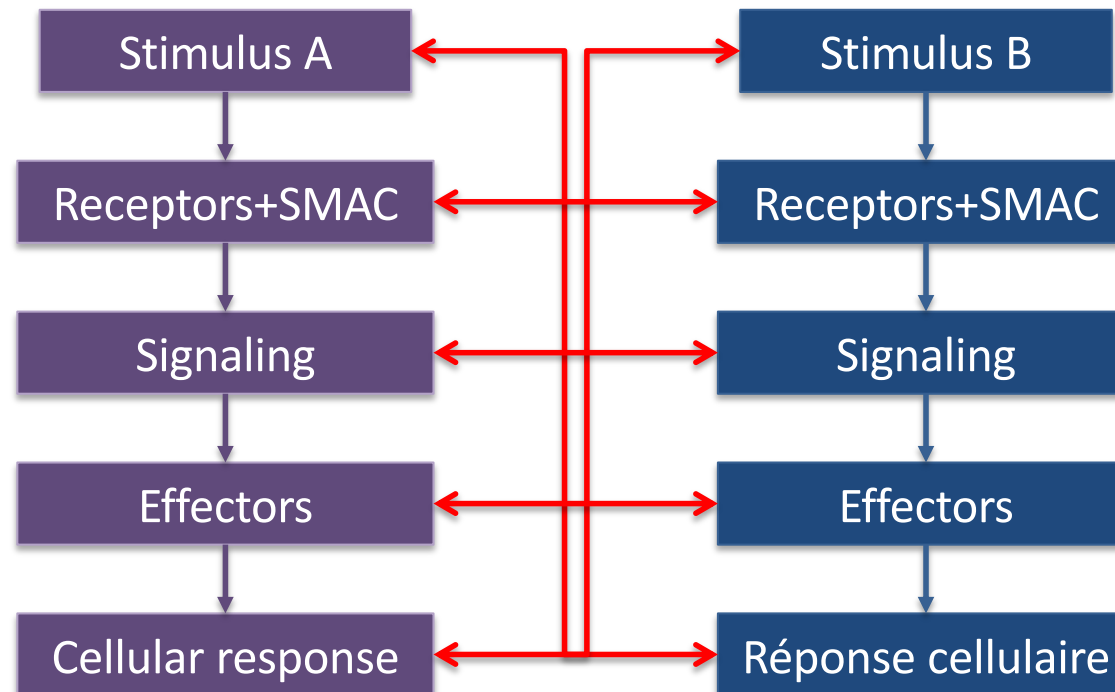
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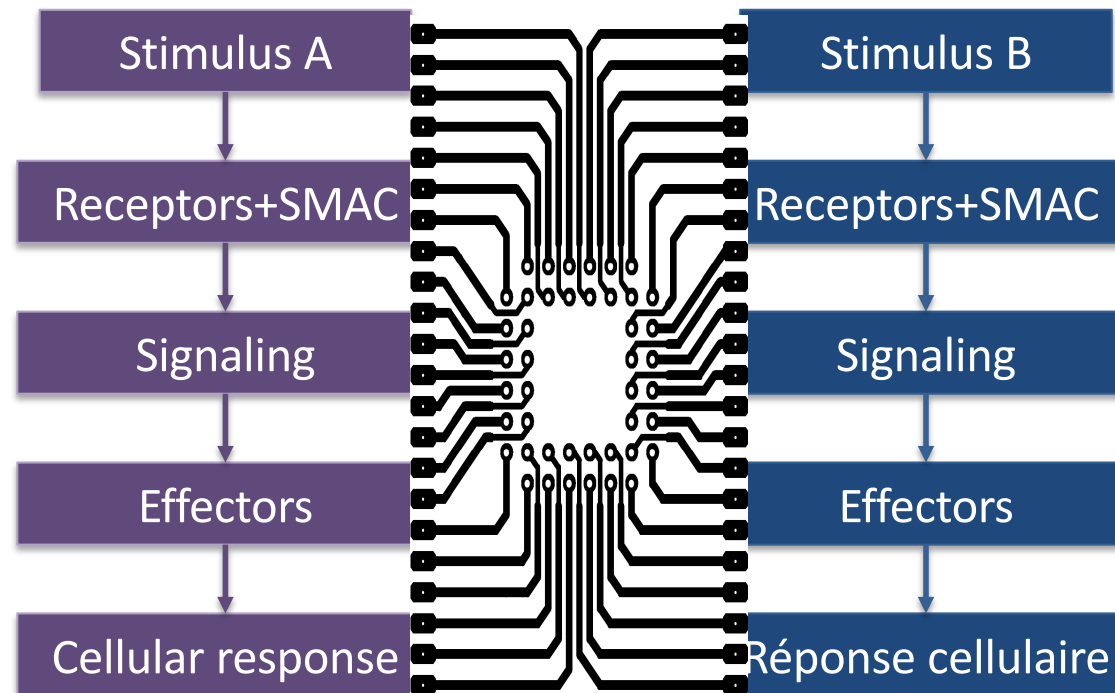
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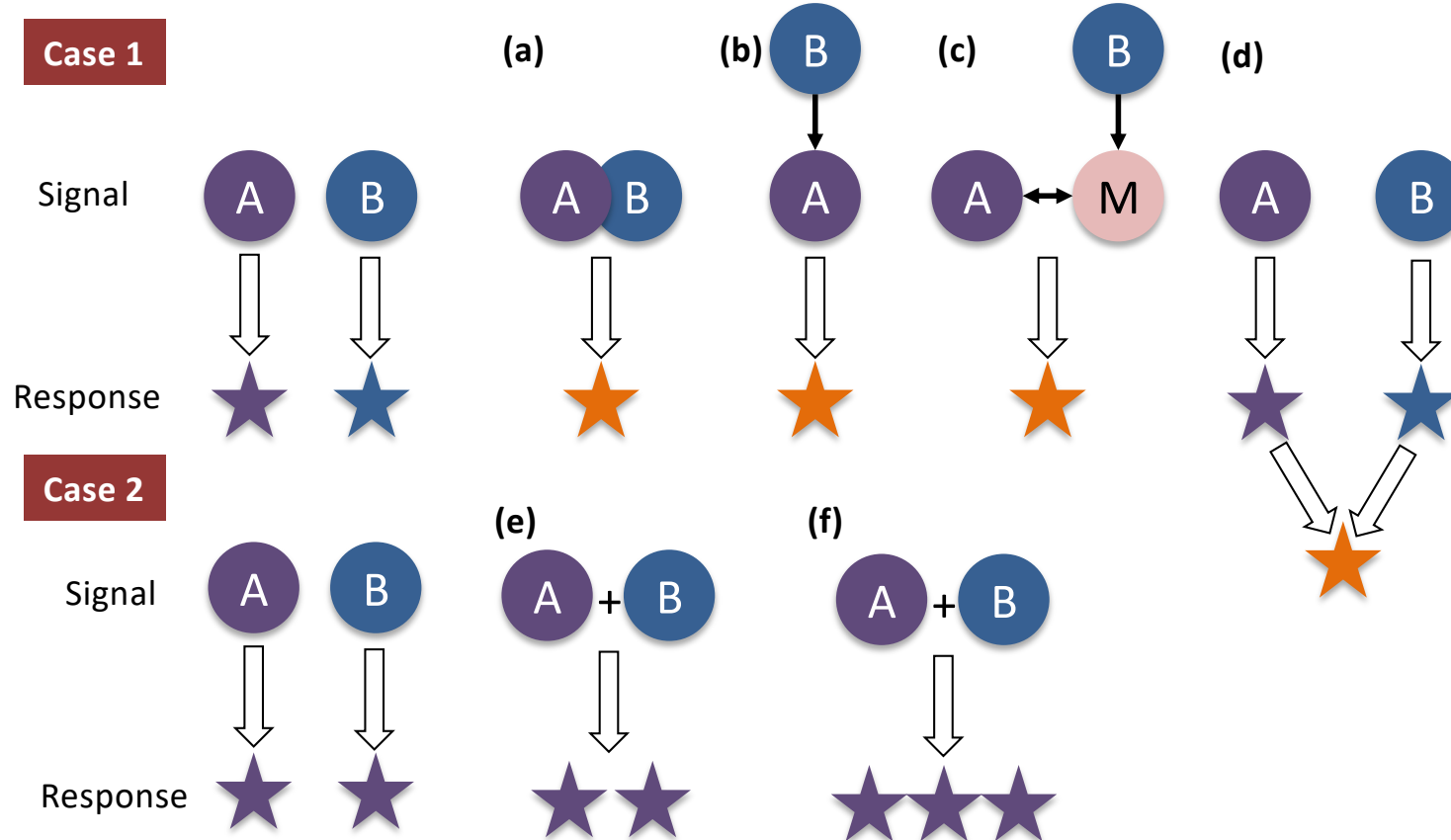
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Several members of these signaling pathways are then usually shared by separate pathways, thus resulting in interactions in between pathways.



What is a cross-talk ?

Cross-talks may take place at any step of a signaling pathway. There is no universal classification of the different types of cross-talks.



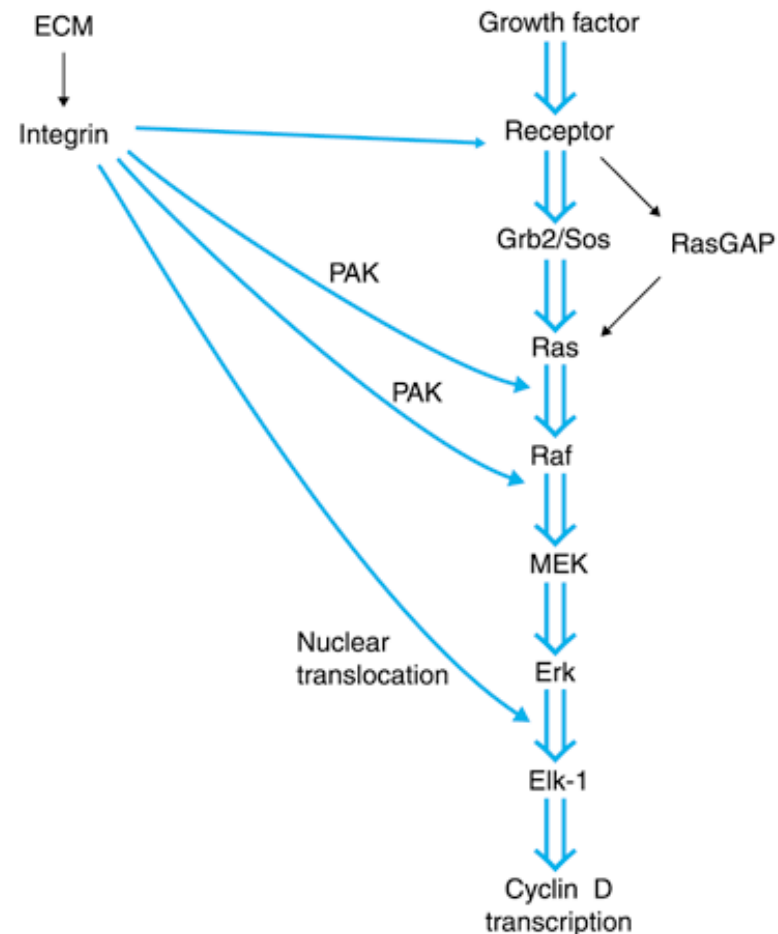
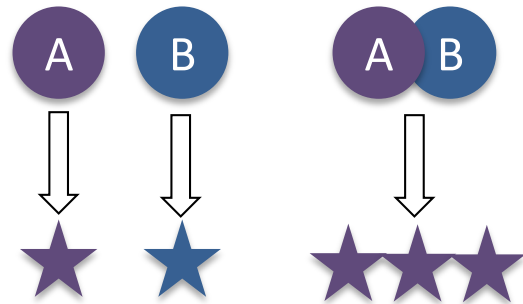
We can define a cross-talk when the stimulation of both signals A+B produces a different response than A alone or B alone (either in the nature or in the amount of the response. Thus, ★ may in fact be ☆☆)

At the molecular level, at least one element of a pathway must be connected to an element of the other pathway:

- Through direct interaction **(a)**
- If the element of a pathway is the enzymatic or transcriptional substrate of the other pathway **(b)**
- If the element of a pathway regulates the modulator of the other pathway **(c)**
- A particular case (d) concerns the regulation of gene expression. The expression of the gene ★ requires two transcription factors ☆ and ☆ without a real cross-talk of pathways A and B. Here, this case is referred to as *cooperation*.

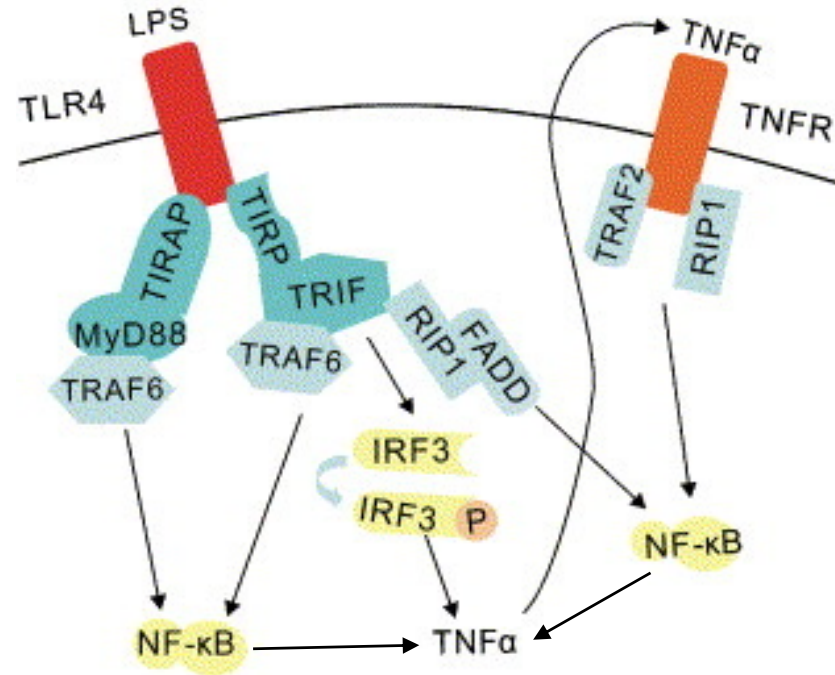
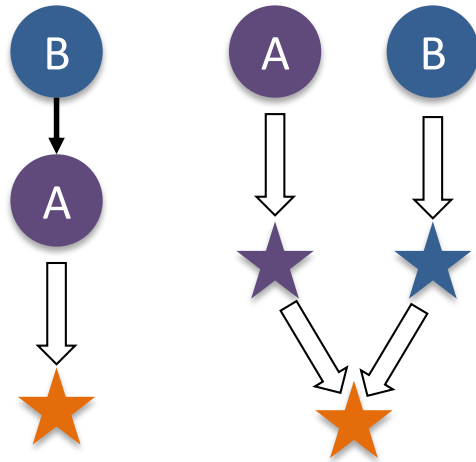
When two pathways lead to a similar result (case 2), and the activation of the two pathways leads to a response corresponding to the sum of both pathways **(e)**, then it is not a cross-talk (*additive effect*). If the result is higher than the sum of the two separate pathways, then the effect is called synergistic but may also be called a cross-talk **(f)**.

Some examples of cross-talks



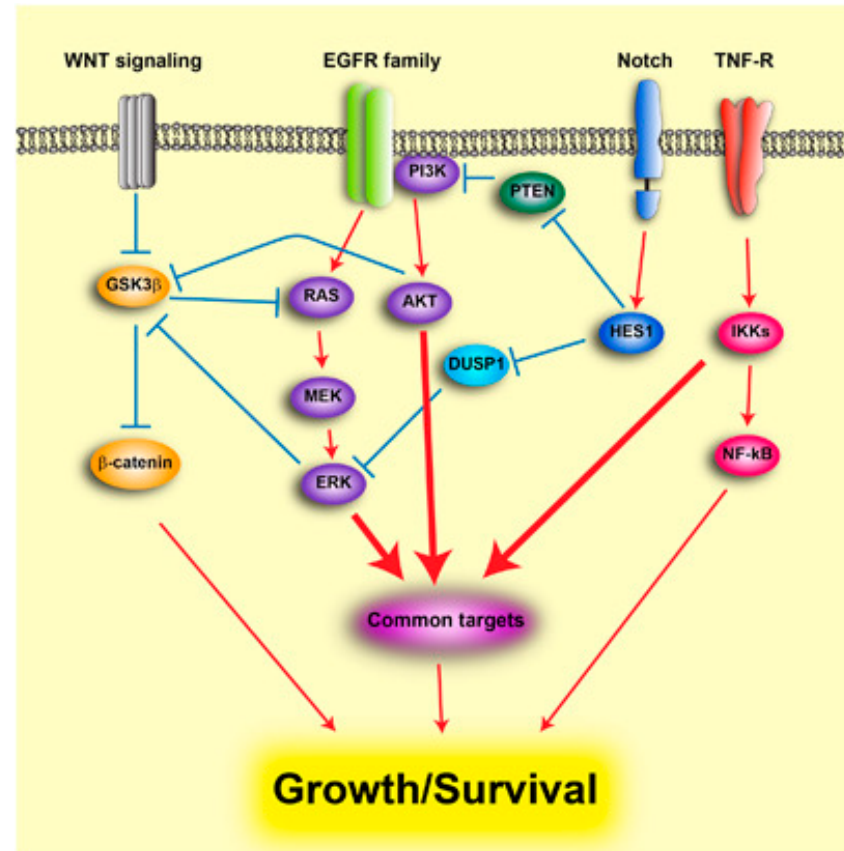
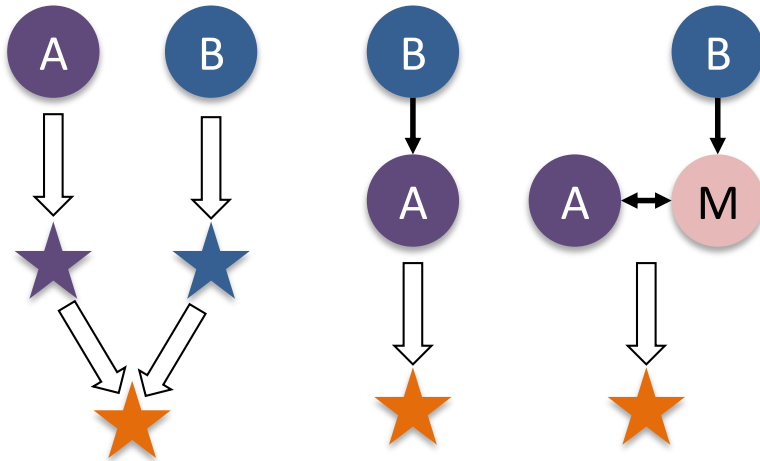
The activation of the MAPK pathway by growth factors, inducing cell proliferation, is amplified by integrins, through the activation of p21 activated kinase (PAK). Schwartz MA and Ginsberg MH. *Nat Cell Biol* 2002.

Some examples of cross-talks



The activation of the TLR4 pathway induces the activation of TNFR, through the regulation of TNF α expression. Bo Zhong et al., Virology 200

Some examples of cross-talks



Cross-talks between the EGFR (Epidermal growth factor receptor) pathway and other pathways.
Yamagushi H et al., Oncogene 2013.

Global analysis of signaling pathways

Natarajan M et al., Nature Cell Biology, 2006.

Aim: To define the coordinated activities of intracellular signaling pathways in response to several stimuli.

Cells: **RAW 264.7** (*Mouse leukaemic monocyte/macrophage cell line*).

Strategy: Systematic measurements of cellular responses induced by treatment with 22 stimulus, alone or in pairs (231 pairs).

Stimulations: Agonists of TLR (toll like receptors), GPCR, cytokines receptor, receptor with tyrosine kinase activity.

Receptors that are ususally co-activated under physiological conditions.

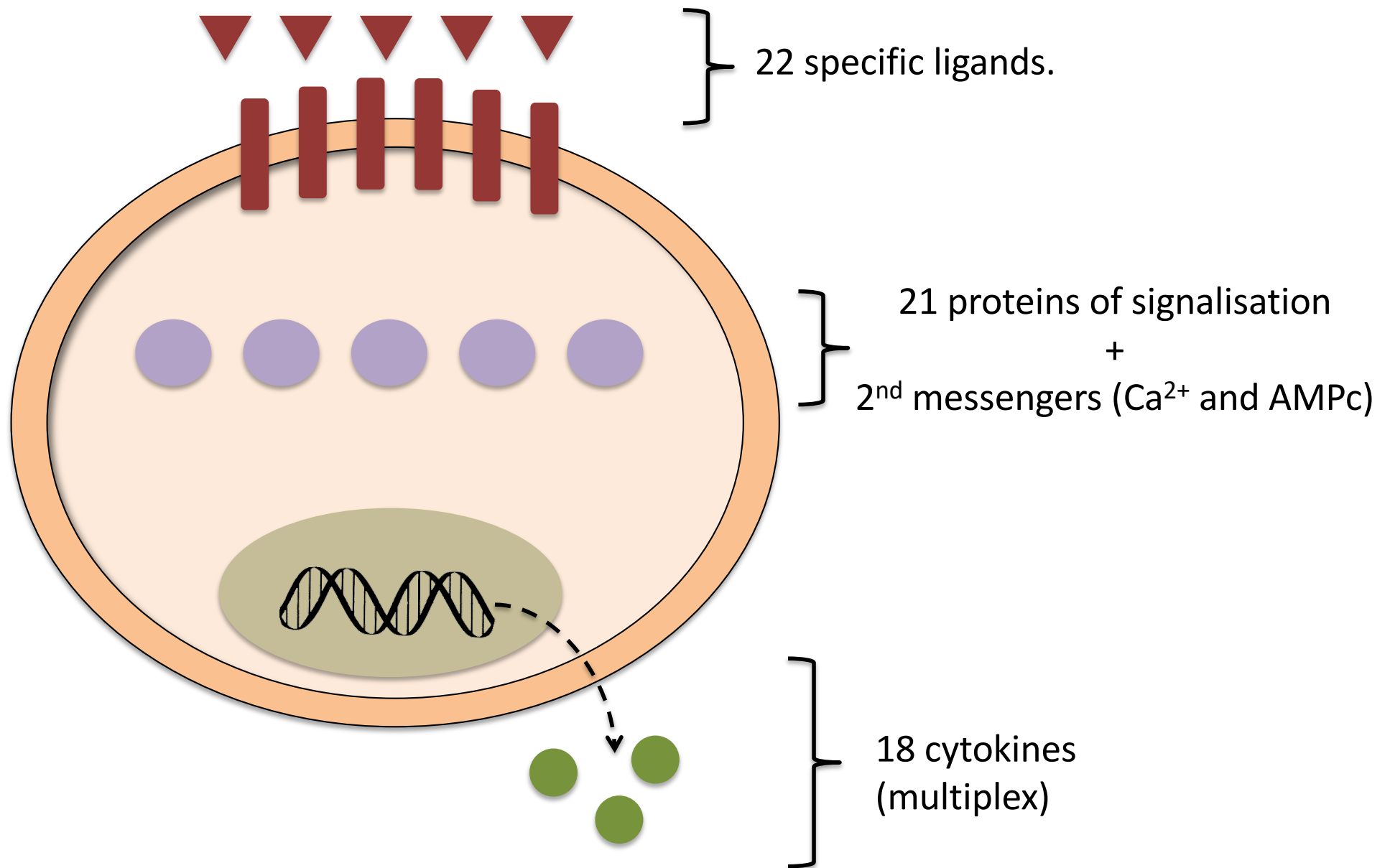
Measured parameters (Readout):

Direct parameter : Secretion of 18 cytokines.

Indirect parameters :

- Ca^{2+} flux
- cAMP production
- Phosphorylation of 21 proteins of signalisation (contained in all signaling modules)

Global analysis of signaling pathways



Global analysis of signaling pathways

Ligands

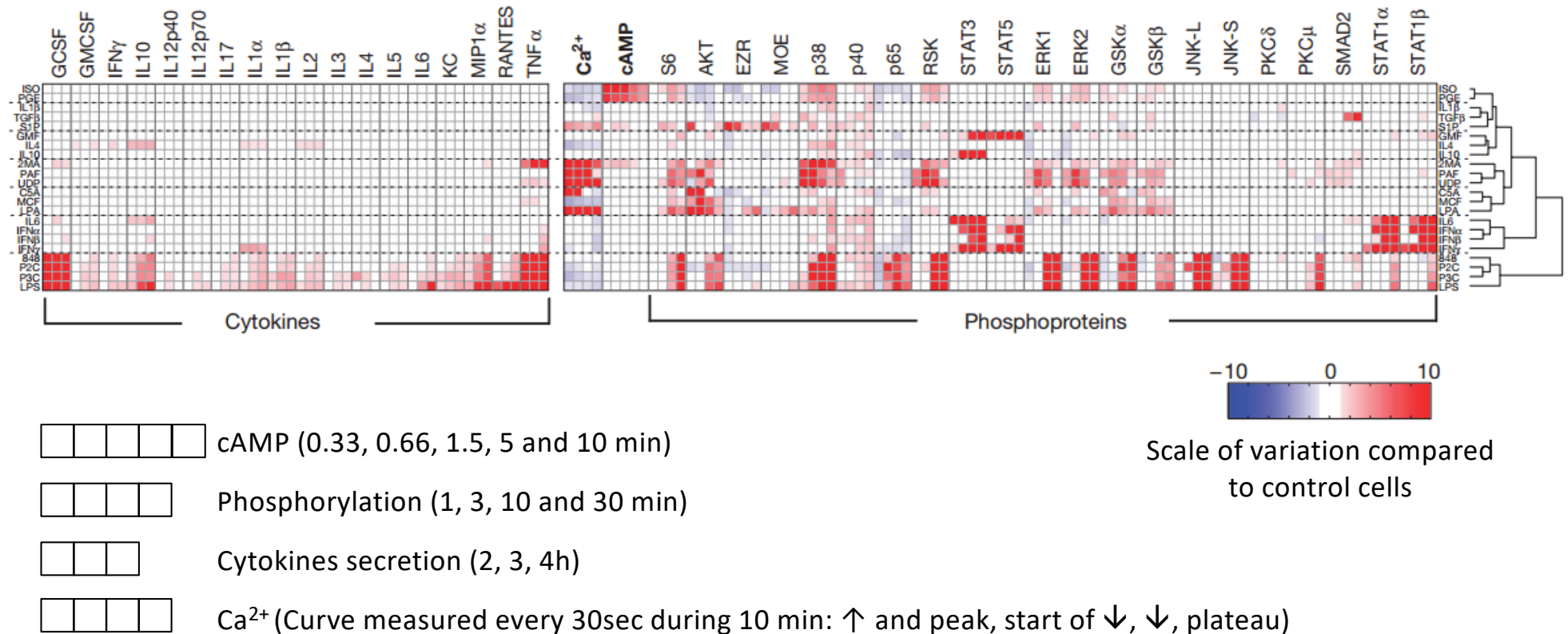
Class	Ligands	Receptors (transducers)
Bacterial and pathogens	LPS, 848, P2C, P3C	TLR 1,2,4,6,7,8 (MyD88)
Purinergic Agonists	2MA, UDP	P2YR (GPCR G α q, Ca ²⁺)
Complement	C5A	C5A-R (GPCR G α i, Ca ²⁺)
Cytokines (CSF)	GMF, MCF	GMF-R (JAK), MCF-R
Cytokines (Interleukines)	IL4, IL6, IL10, IL1 β	IL-R (divers)
Cytokines (Interferons)	INF α , INF β , INF γ	INF-R (JAK)
Cytokines	TGF β	TGF-R (Smad)
Phospholipids	LPA, SIP	EDGR (GPCR G12/G13 ? Gq? , Ca ²⁺)
Prostaglandins	PGE	EPR (GPCR G α s)
Lipids	PAF	PAF-R (GPCR G α q, Ca ²⁺)
β -Adrenergic agonist	Isoprénaline	β -AR (G α s)

Global analysis of signaling pathways

Read-outs

Class	Targets (signaling pathways)
Cytokines	GCSF, GMCSF
	IL1 α , 1 β , 2, 3, 4, 5, 6, 10, 12p40, 12p70, 17
	INF γ , TNF α
	MIP1 α , RANTES, KC(CXCL-1)
2 nd messengers	AMPc
	Ca ²⁺
Phospho-proteins	S6, RSK (ribosomal 40S)
	AKT, GSK α , GSK β (Pi3K)
	p38, ERK1, ERK2, JNK-I, JNK-S (MAPK/ERK)
	P65 (NF-kB)
	PKC δ , PKC μ (DAG)
	STAT1 α , STAT1 β , STAT3, STAT5 (JAK)
	EZR, Moesin (phosphatidylinositol)
	P40 phox (NADPH oxidase subunit)
	Smad2 (TGF-R)

Profile of responses after simple stimulations



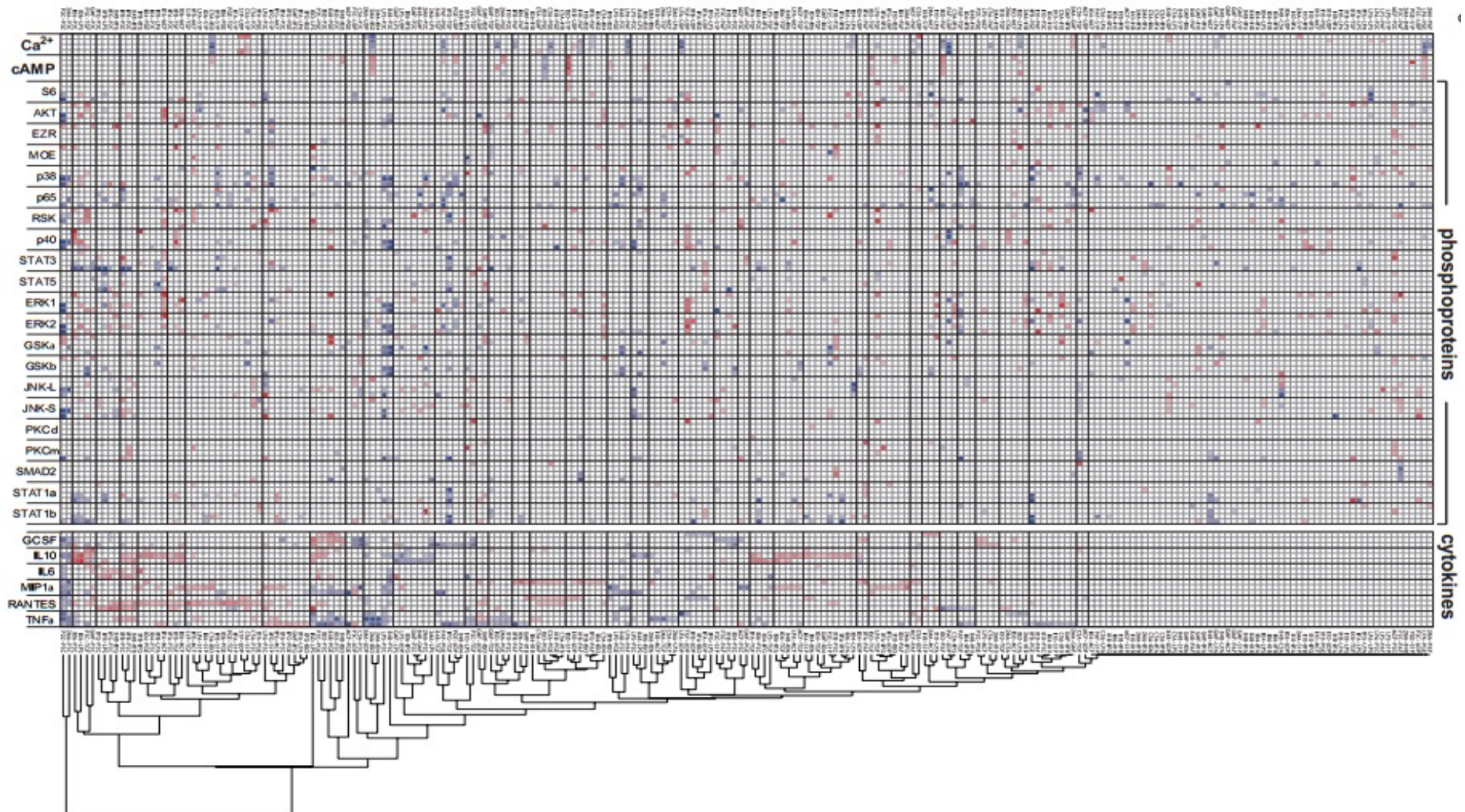
The profile of responses following stimulation with one ligand fits perfectly with the known signaling mechanisms and allows to sort ligands into functional groups (-----). This results validates the analytical tool developed to present experimental data in simple profiles of responses.

Profile of responses after dual stimulations

- Ligand A $\Rightarrow$ Response A and Ligand B $\Rightarrow$ Response B
- Ligand A + Ligand B $\Rightarrow$ Response A + Response B
- Ligand A + Ligand B $\Rightarrow$ Response C then A and B interact to provide a response C $\Rightarrow$ **Cross-talk**
- If A and B activate the same pathway and if the dual stimulation provides a higher activation level than the sum of each response, then the effect is synergistic, a special form of cross-talk.

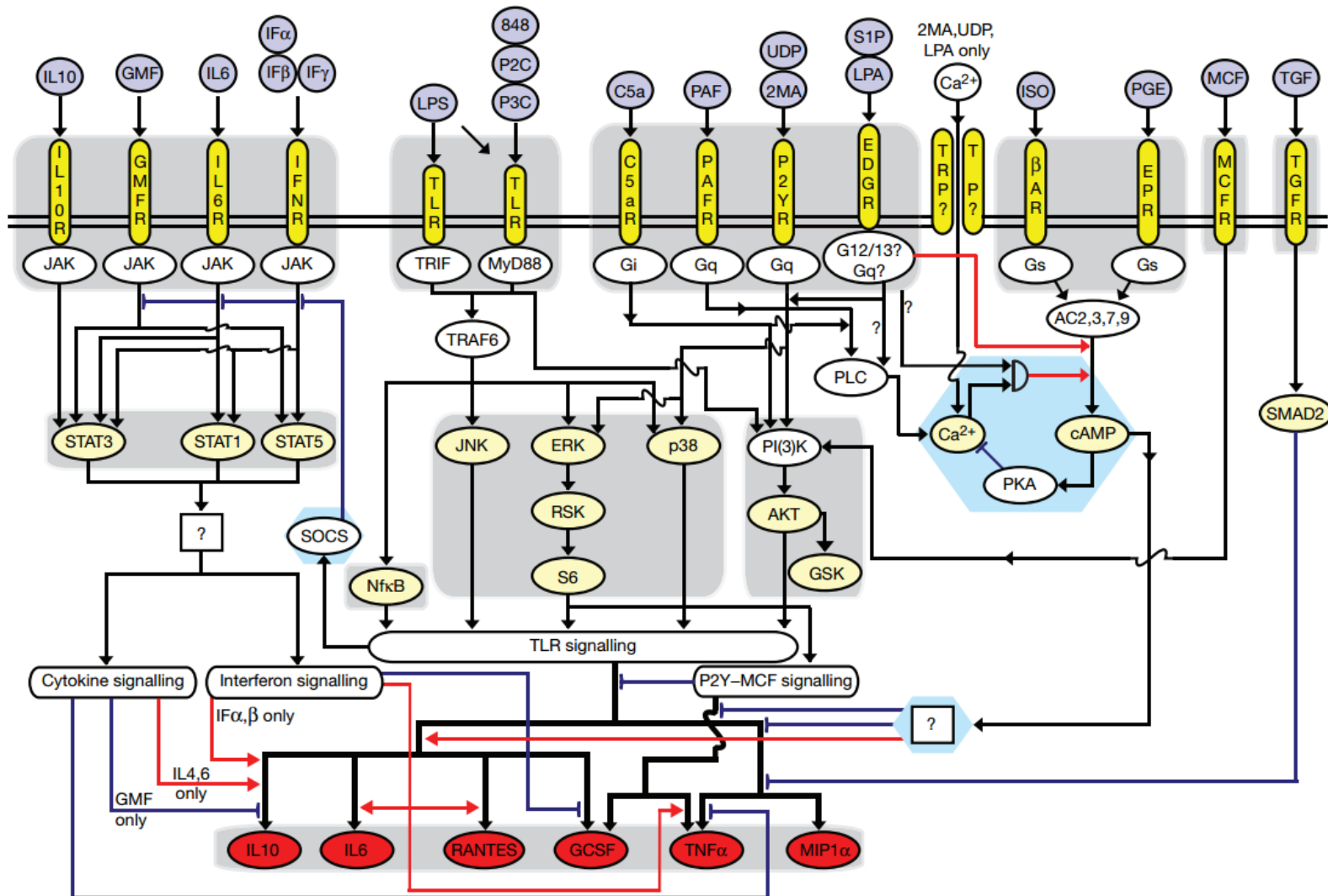
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Profile of responses after dual stimulations

The analysis of data and classification of additive effects depending on the class of receptors and the identified interactions allowed to propose a model in the specific cell type:



NF- κ B transcription factors



David Baltimore



Rajan Sen

“Early in the course of writing the first manuscript describing immunoglobulin gene enhancer-binding proteins (1986), we came to a point at which David Baltimore said to me, “You have to give this thing a name; it will stick to you for the rest of your life.” He was referring to a B lymphocyte-specific immunoglobulin κ -chain enhancer-binding protein that we had identified; we named it NF- κ B to highlight its properties. We now know that this factor is not limited to B cells. Instead, NF- κ B is involved in many important biological functions in almost every cell type examined so far.”

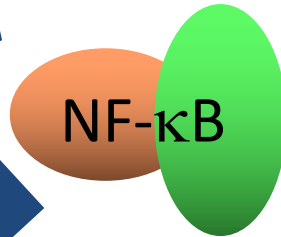
Rajan Sen, Nat Immunology. 2011

NF- κ B transcription factors

STIMULI

- Physiological mediators
- Viral proteins
- Bacterial proteins
- Cytokines
- Ligands
- Stress

- p50
- p52
- p65 (Rel A)
- Rel B
- c-Rel



GENE REGULATION

- Receptors
- Adhesion Molecules
- Cytokines, chemokines
- Growth Factors
- Transcription Factors

CELLULAR RESPONSES

- Immune Response
- Inflammatory response
- Proliferation
- Anti-apoptosis Mechanisms

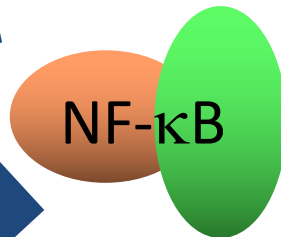


NF-κB transcription factors

STIMULI

- Physiological mediators
- Viral proteins
- Bacterial proteins
- Cytokines
- Ligands
- Stress

- p50
- p52
- p65 (Rel A)
- Rel B
- c-Rel



ASSOCIATED PATHOLOGIES

- Leukaemia, lymphoma
- Cancers
- Inflammatory diseases

Dysregulations

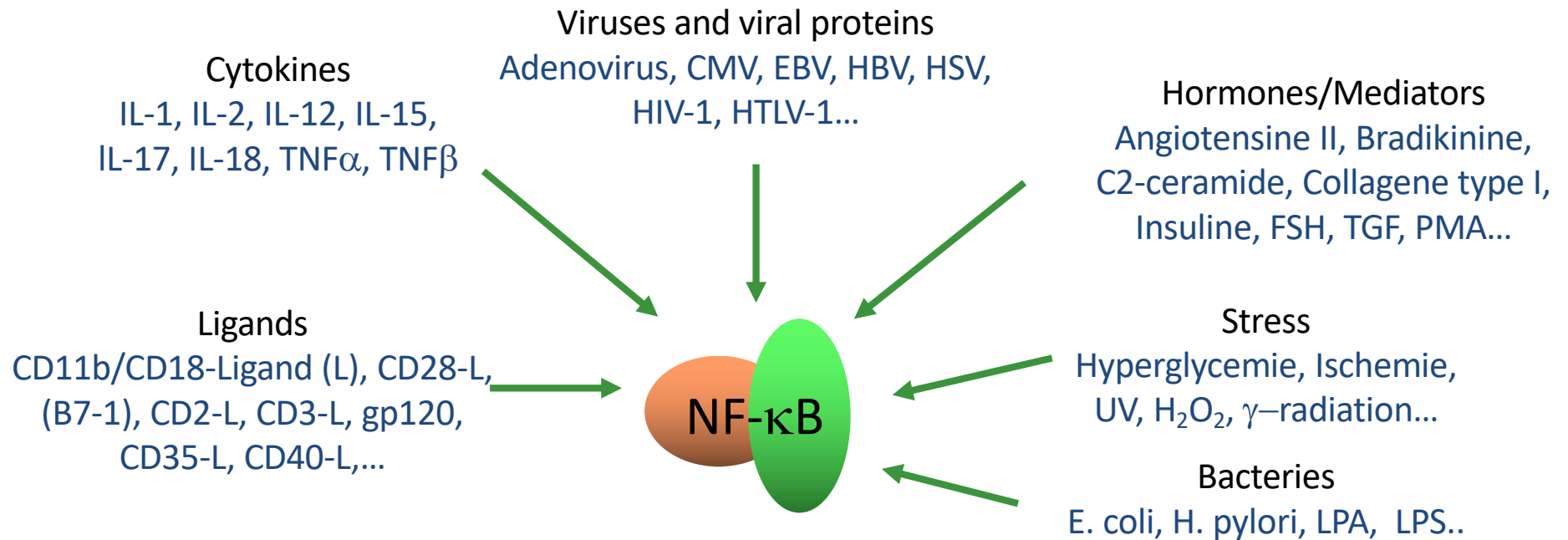
GENE REGULATION

- Receptors
- Adhesion Molecules
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- Growth Factors
- Transcription Factors

CELLULAR RESPONSES

- Immune Response
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Activators of NF- κ B



Target genes of NF-κB

Cytokines/chemokines

IL-1, IL-2, IL-6, IL-8, IL-11, IL-15, Eotaxin, IFN α , IL1 β , IP-10, KC, Lymphotoxin α Lymphotoxin β , MCP-1, MIP1a,b, MIP-2, RANTES, TNF α , TNF β

Acute phase Proteins

Angiotensinogen, C4b binding Protein, Complement factor B, Amyloid A precursor, TF-1

Immunoreceptors

IL-2R α -chain, Ig C γ 1, Ig ϵ hc, Ig κ lc, H-2K, HLA-B7, β 2-Microglobulin, B7.1 (CD80), BRL-1, CCR5, CD23, CD48, TCR β chain

Adhesion Molecules

ELAM-1, ICAM-1, madCAM-1
P-selectin, Tenascin-C

Stress response Genes

COX-2, 12-Lipoxygenase, Inducible NO-synthase, Mn SOD

Regulators of apoptosis

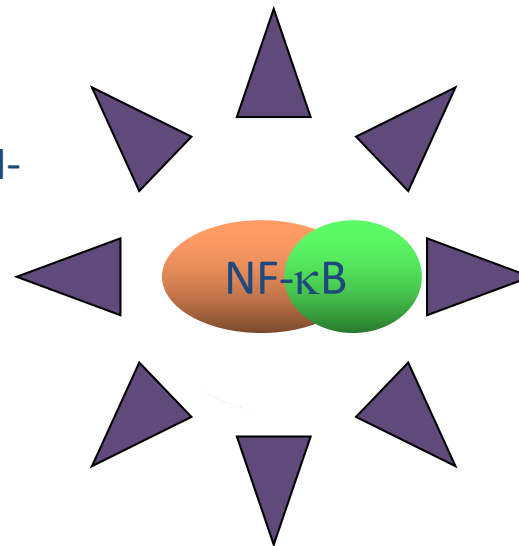
Bcl 1/A1, Bcl-xL, CD95, c-FLIP, Fas-ligand, IAPs

Transcription Factors

c-rel, nfkb1, nfkb2, I κ B α , I κ B ϵ , junB, c-myc, c-myc, IRF-1, IRF-2, p53

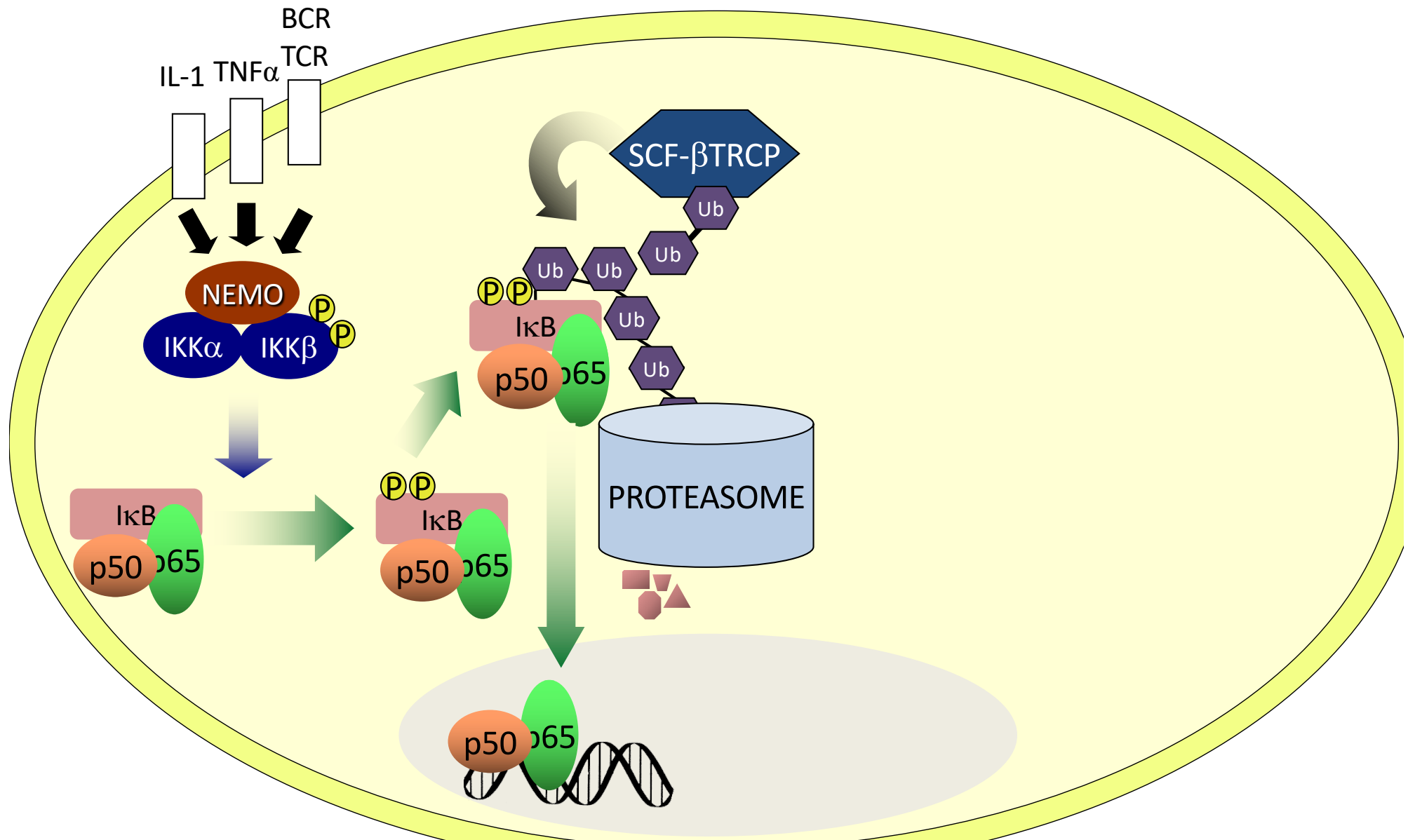
Growth Factors

G-CSF, GM-CSF, IGFBP-2, M-CSF, PDGF B chain, proenkephalin, VEGF



NF- κ B activation mechanisms

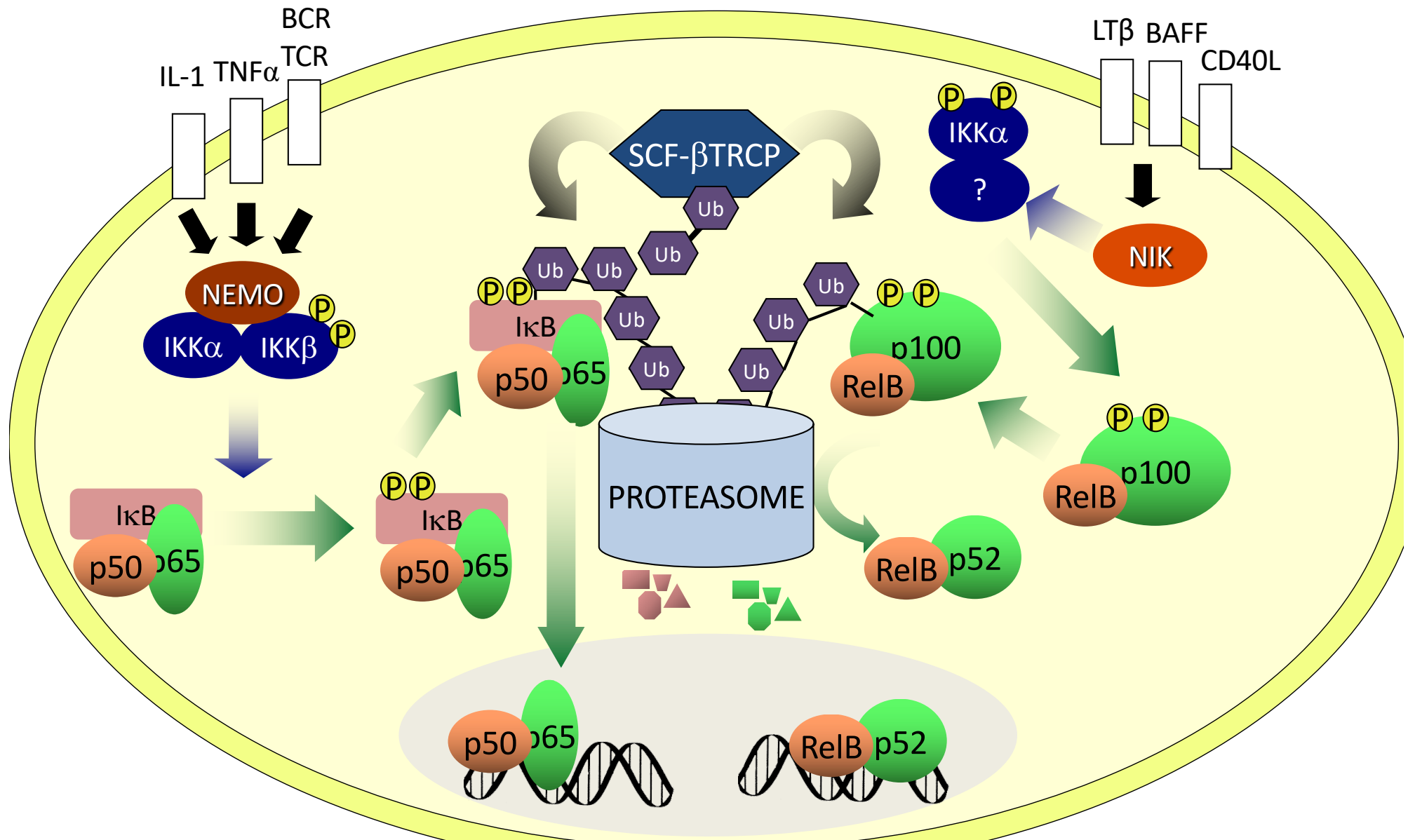
CANONICAL PATHWAY



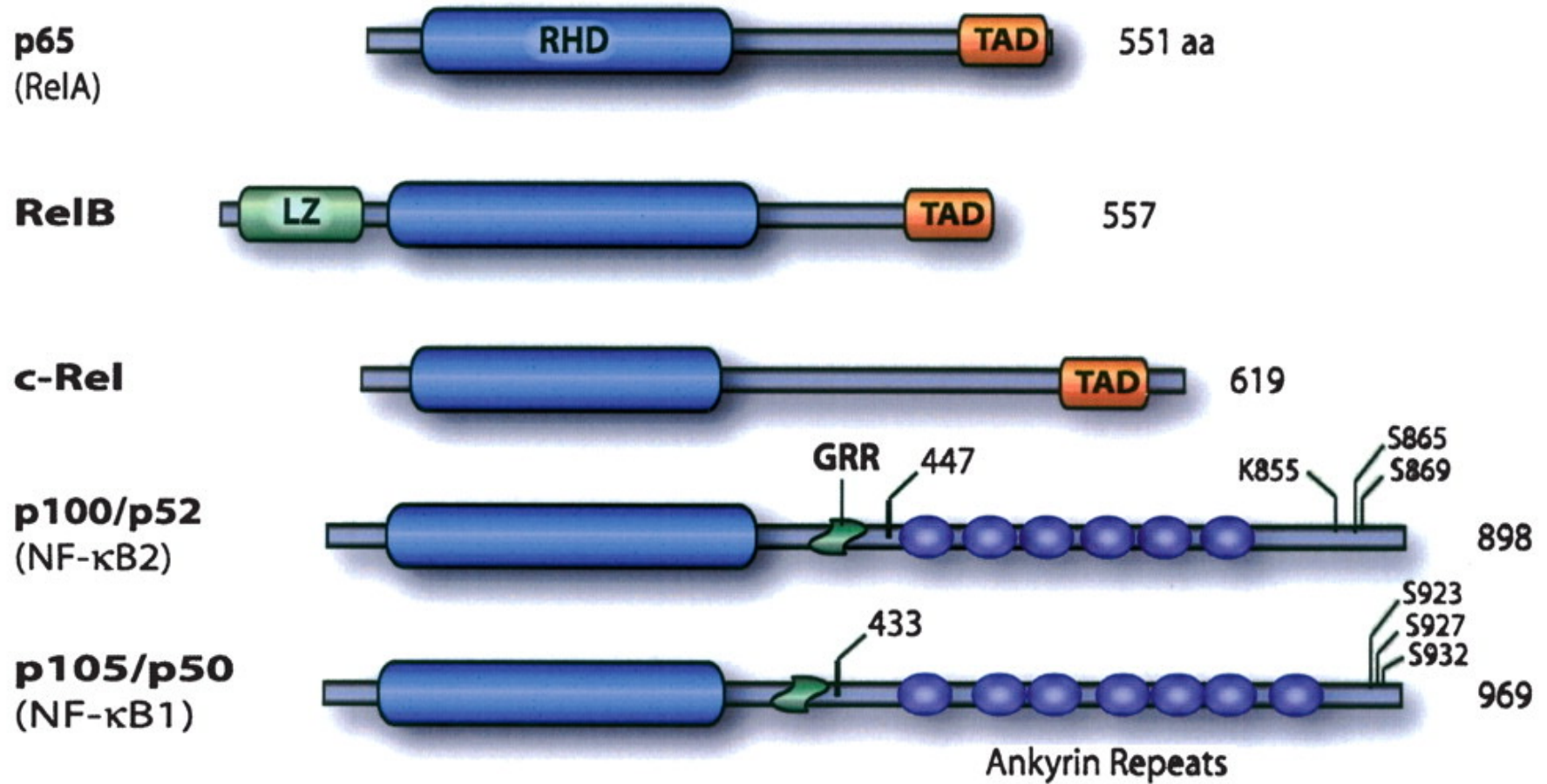
NF- κ B activation mechanisms

CANONICAL PATHWAY

NON CANONICAL PATHWAY



Members of the NF- κ B family: transcriptional activators



RHD: Rel Homology Domain

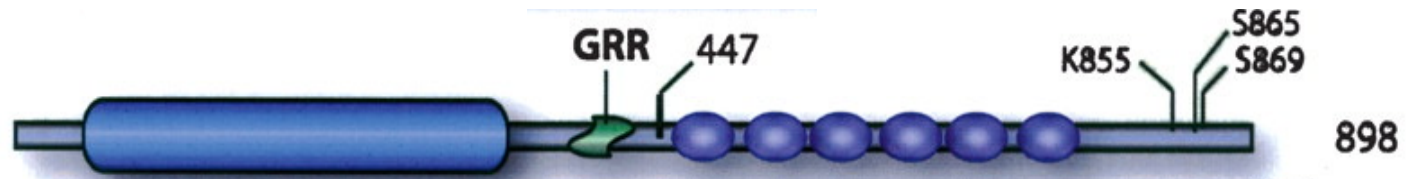
TAD: Transactivator Domain

LZ: Leucine Zipper

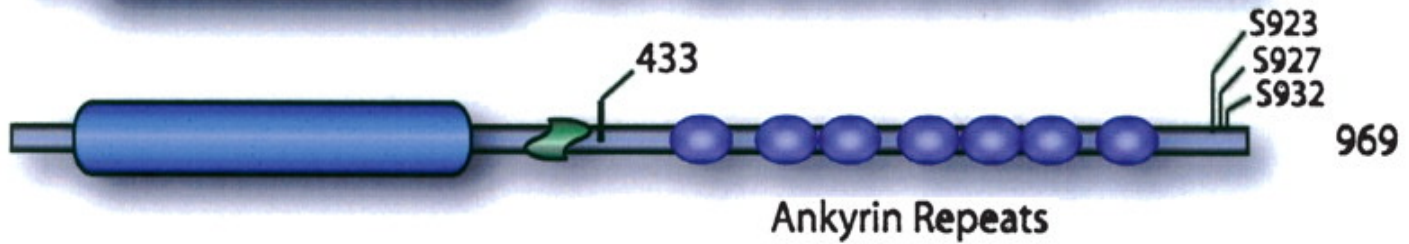
GRR: Glycin-Rich Region

Members of the NF- κ B family : inhibitors of I κ B

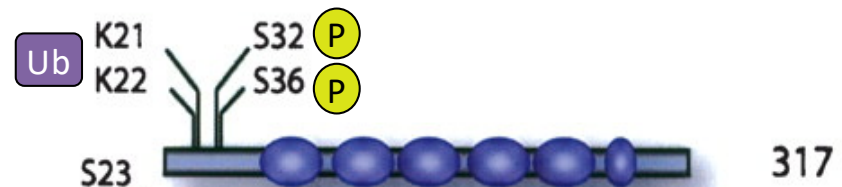
p100/p52
(NF- κ B2)



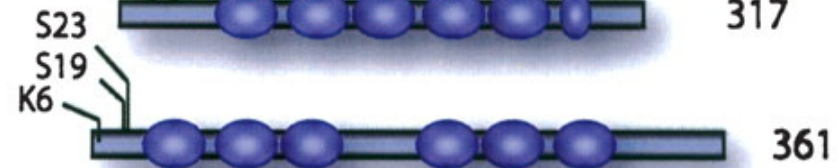
p105/p50
(NF- κ B1)



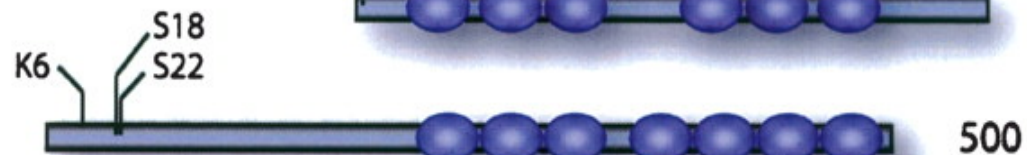
I κ B α



I κ B β



I κ B ϵ



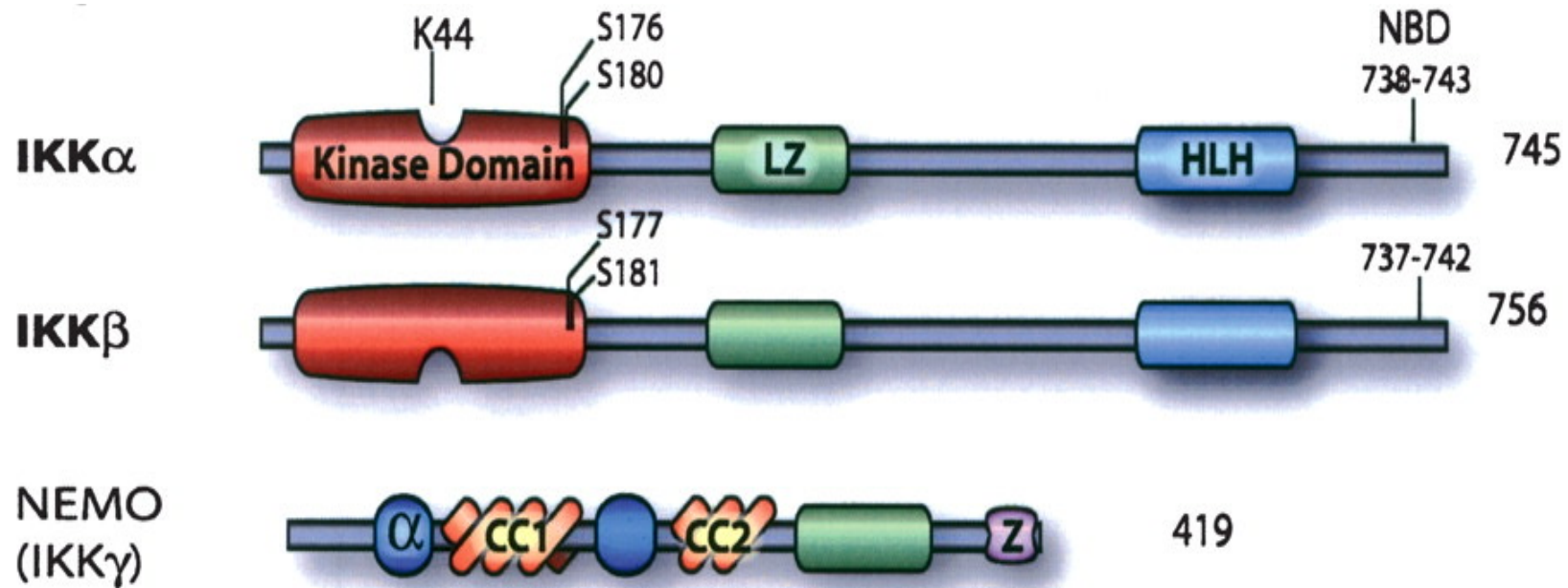
I κ B γ



BCL-3



Members of the NF- κ B family : IKK regulators



HLH: Helix-Loop-Helix

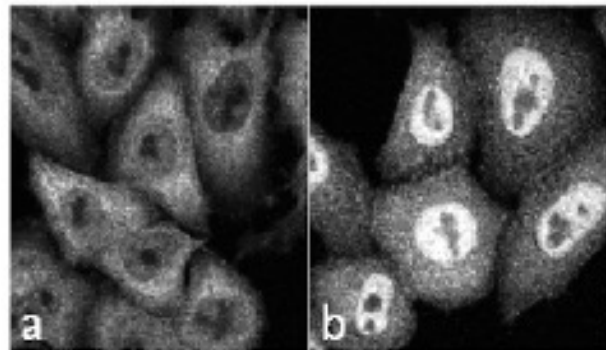
CC: Coiled-Coil

LZ: Leucine Zipper

GRR: Glycin-Rich Region

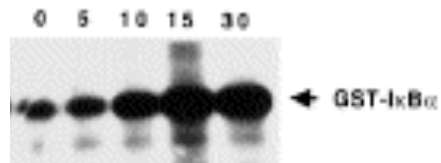
Techniques to determine and measure the activation of NF- κ B

Nuclear translocation of p65

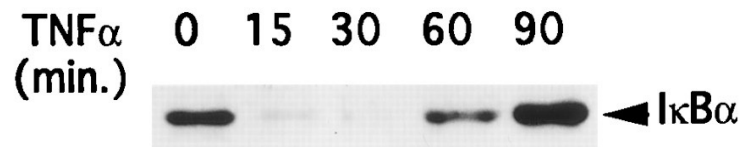


-TNF α

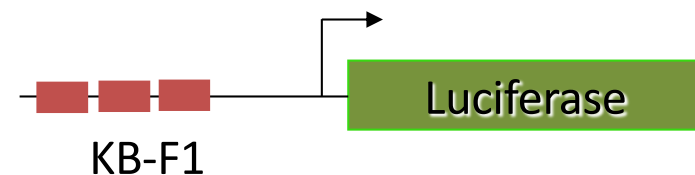
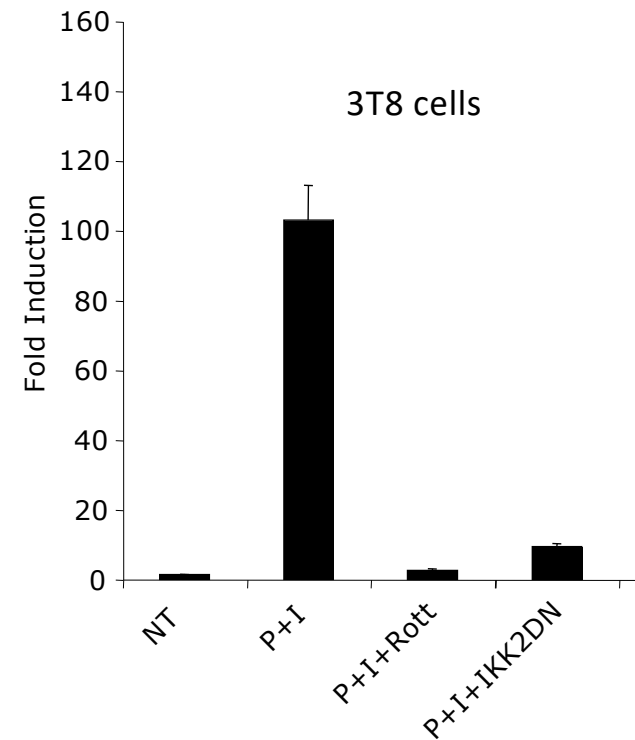
+TNF α



Phosphorylation of I κ B α

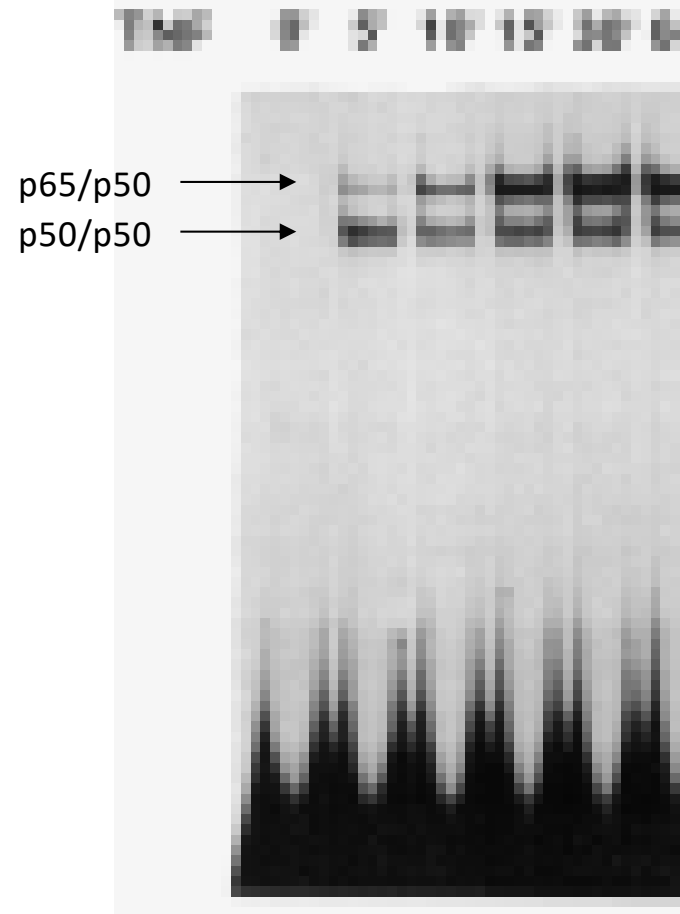
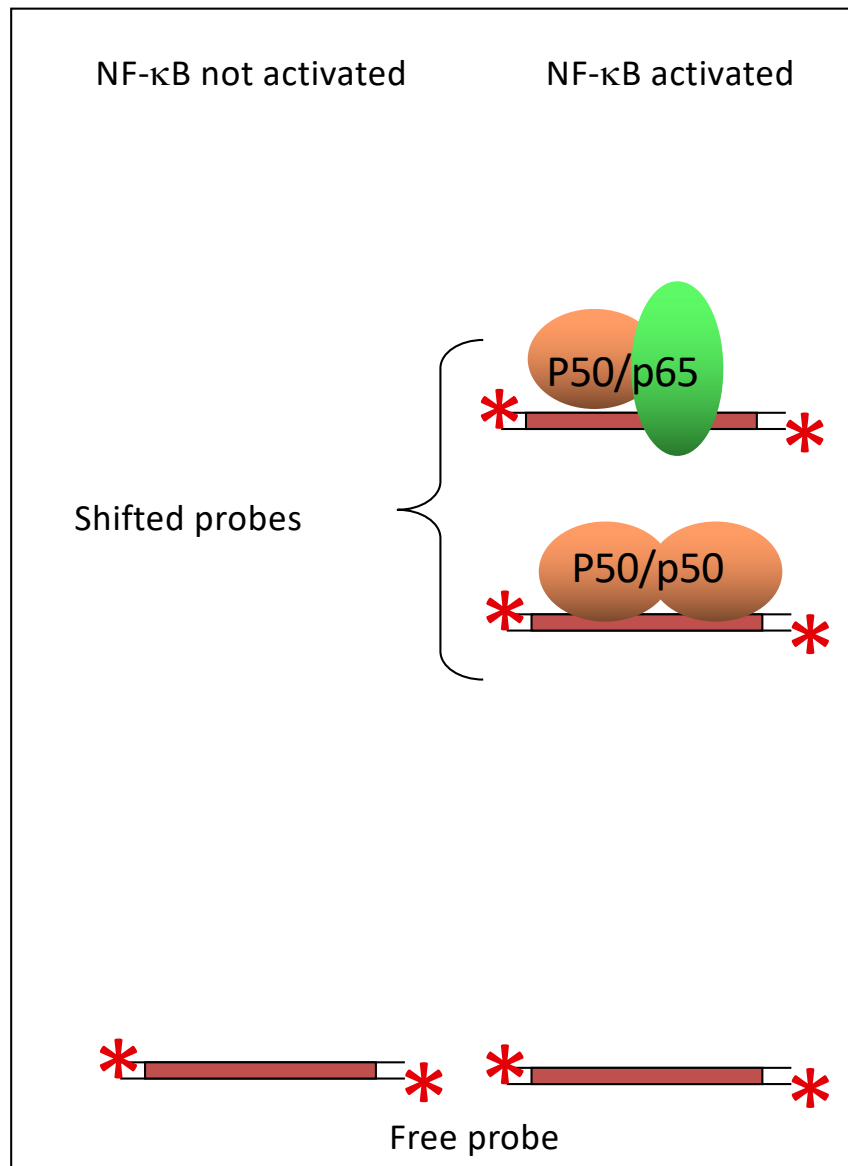


Degradation and re-synthesis of I κ B α



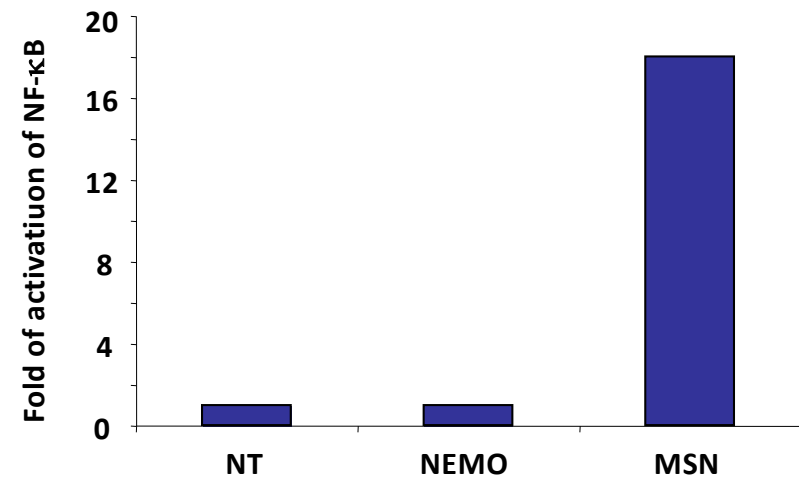
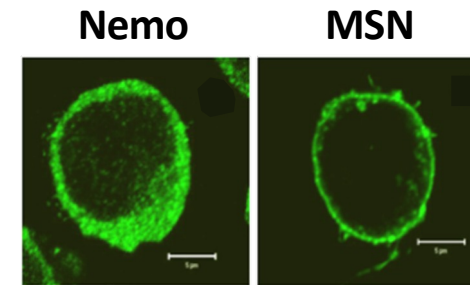
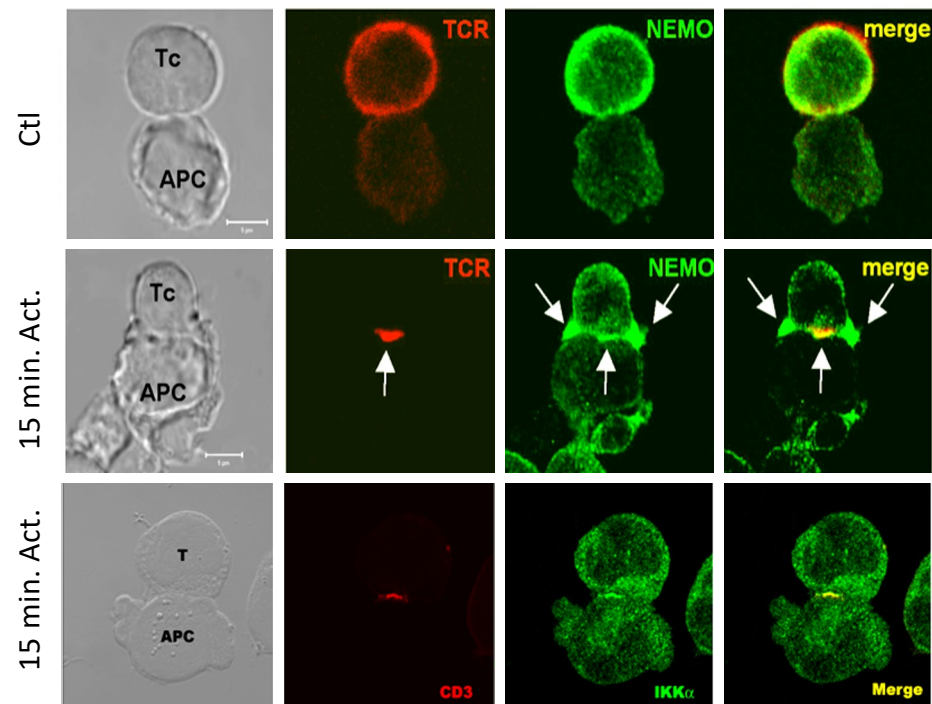
Activity of a reporter gene NF- κ B

Techniques to determine and measure the activation of NF- κ B



Binding of NF- κ B to DNA: Gel Shift

Activation of NF- κ B relies on the relocalization of the complex NEMO/IKK



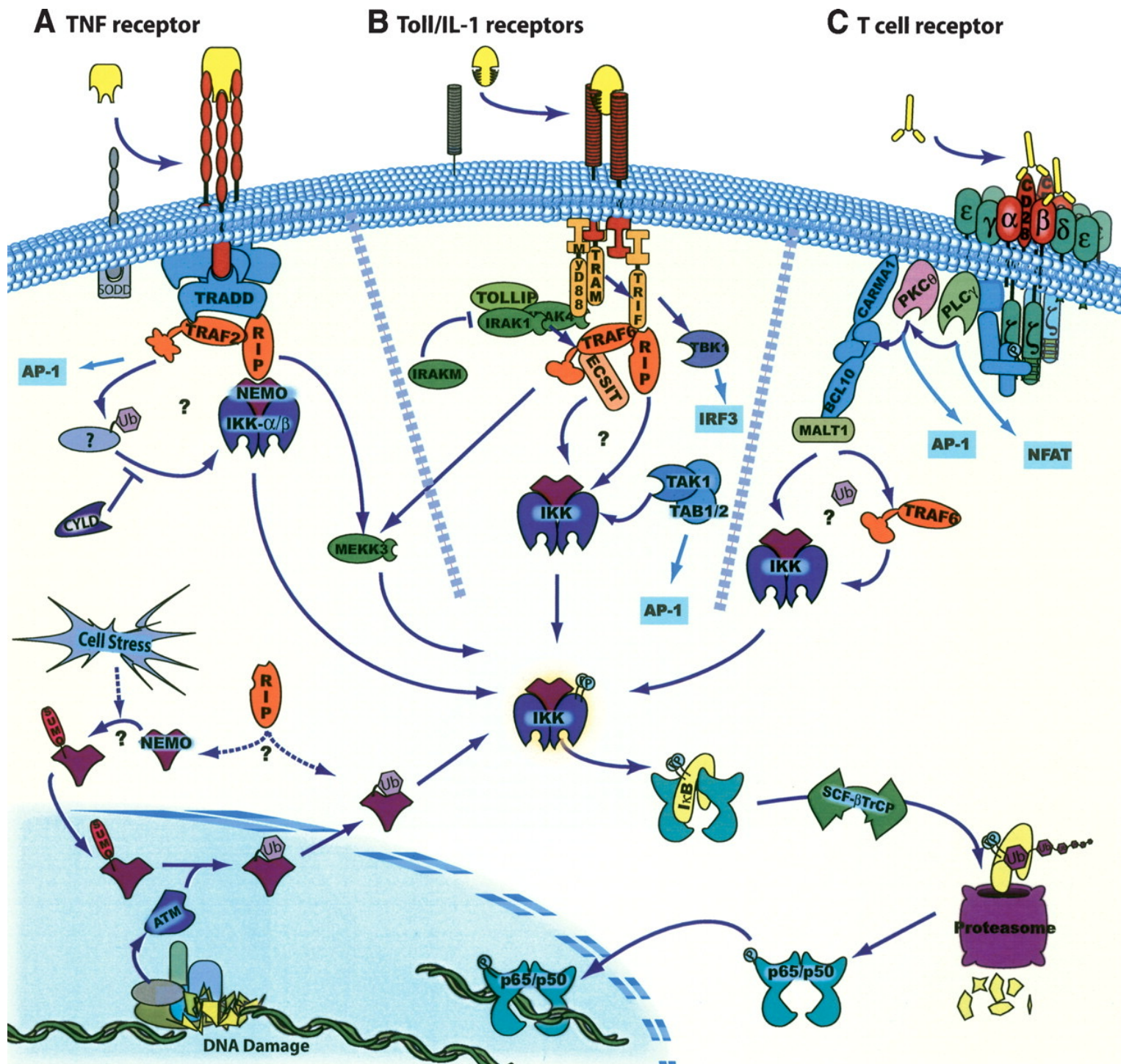
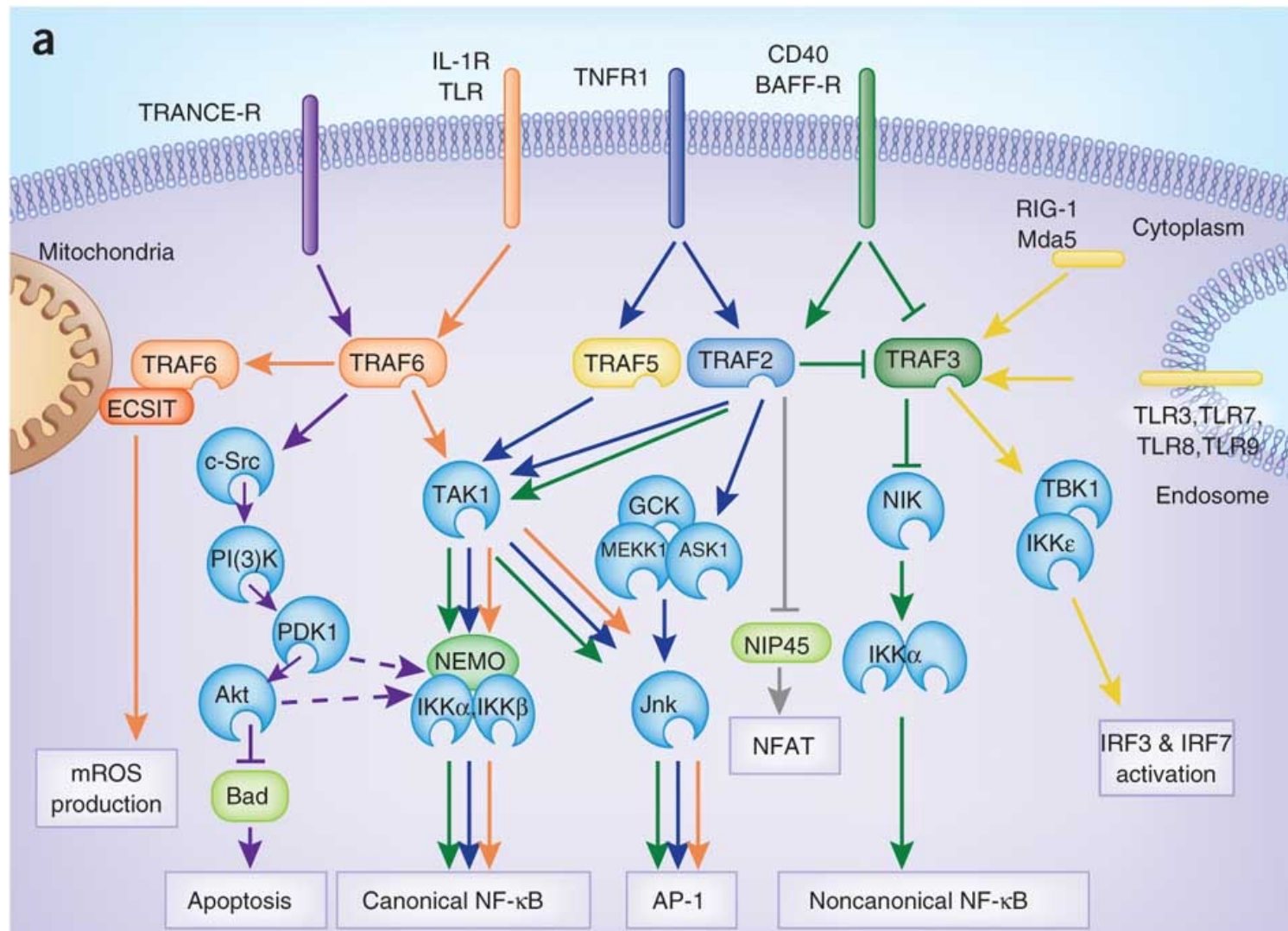


Table 1. Kinases known to phosphorylate the NF- κ B and I κ B proteins^{a,b}

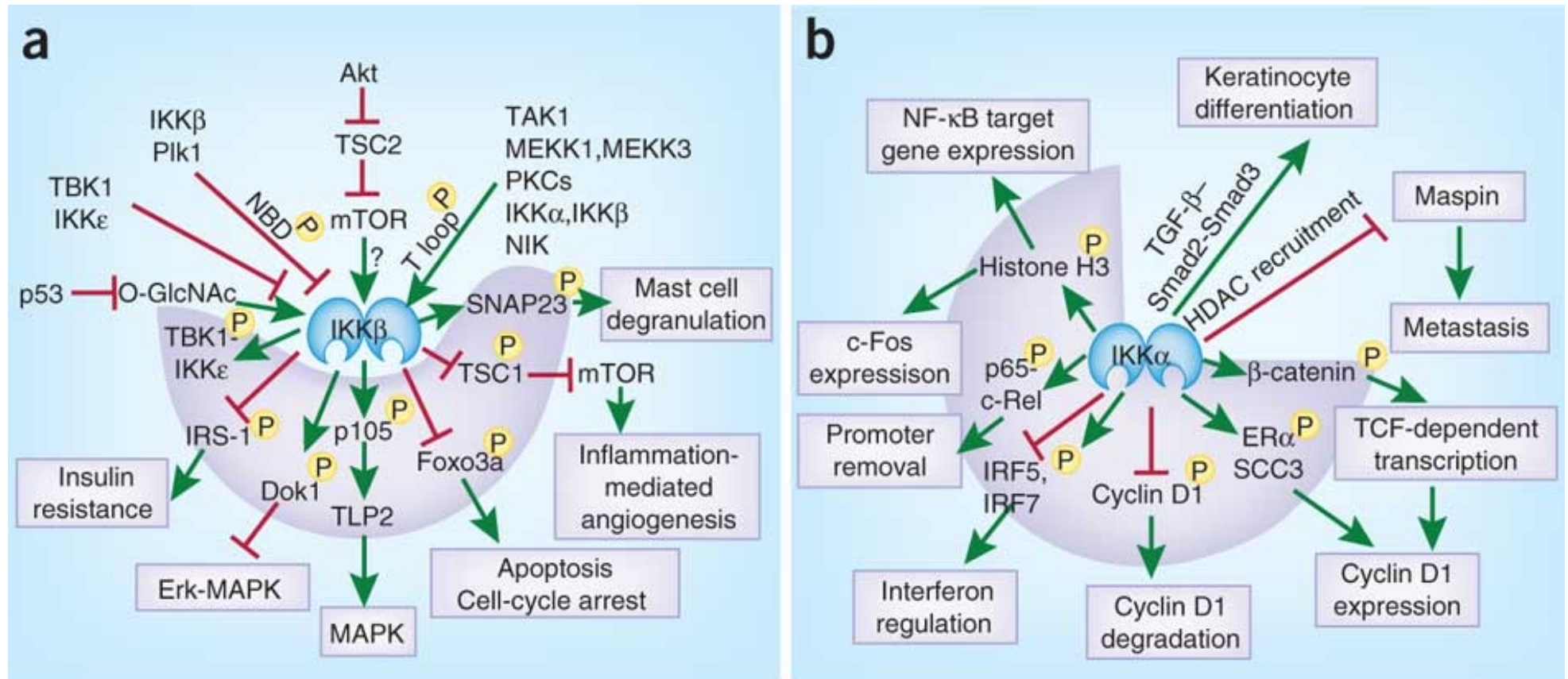
Substrates	Kinases	Residues	Location	Function	Biological stimuli
I κ B α	IKK β	Ser32, Ser36	N-terminal domain	Proteasome-mediated degradation	TNF α , IL-1 β
	CK2	Ser293	PEST domain	Destabilization	Constitutive
	Syk	Tyr42	N-terminal domain		H ₂ O ₂
	CK2	Ser283–Thr299	PEST domain	Degradation	UV light
I κ B β	IKK β	Ser19, Ser23	N-terminal domain	Proteasome-mediated degradation	TNF α , IL-1 β
I κ B γ	IKK β	Ser18, Ser22	N-terminal domain	Proteasome-mediated degradation	TNF α , IL-1 β
p100	IKK α	Ser108, Ser115, Ser123, Ser872	N- and C-terminal domains	Processing into p52	CD40, BAFF, lymphotaxin β
p105	IKK β	Ser927, Ser932	PEST domain	Processing into p50	TNF α
	Tpl2	Indirect phosphorylation		Processing into p50?	Unknown
	GSK3 β	Ser903, Ser907		Stabilization	Constitutive
BCL-3	GSK3	Ser394, Ser398	C-terminal domain	Proteasome-mediated degradation	Constitutive
RelA/p65	PKAc	Ser276	Rel homology domain	Regulation of DNA-binding and oligomerization	LPS
	MSK1/2	Ser276	Rel homology domain	Enhanced transactivation potential	TNF α
	PKC ζ	Ser311	Rel homology domain	Enhanced transactivation potential	TNF α
	Unknown	Ser468		Unknown	T-cell co-stimulation (CD3/CD28)
	CK2	Ser529	C-terminal TAD	Enhanced transactivation potential	TNF α , IL-1 β
	IKK α	Ser536	C-terminal TAD	Enhanced transactivation potential	HTLV-1 infection, lymphotaxin β
	IKK β	Ser536	C-terminal TAD	Enhanced transactivation potential	TNF α , T-cell co-stimulation (CD3/CD28)
	Akt	Ser536	C-terminal TAD	Enhanced transactivation potential	IL-1 β
	RSK1	Ser536	C-terminal TAD	Decreased affinity to I κ B α	DNA-damaging agents
	GSK3 β	Four sites within amino acids 354–551	C-terminal TAD	Enhanced transactivation potential	Constitutive
	TBK1	Ser536	C-terminal TAD	Enhanced transactivation potential	IL-1 β
	IKK γ	Ser536	C-terminal TAD	Enhanced transactivation potential	IL-1 β
	Unknown	Ser368	Rel homology domain	Dimerization and p100 stabilization	
	Unknown	Thr84, Ser552		Degradation	T-cell co-stimulation (CD3/CD28), TPA/ ionomycin
Crel	PI3K/PKC	Ser471	C-terminal TAD	Enhanced transactivation potential	TNF α
	PKA-C β	Unknown	C-terminal TAD	Enhanced transactivation potential	Unknown
	Unknown	Unknown tyrosine			G-CSF
p50	PKAc	S337	Rel homology domain	Enhanced binding to DNA	

Cross-talk of the NF- κ B pathway with other pathways



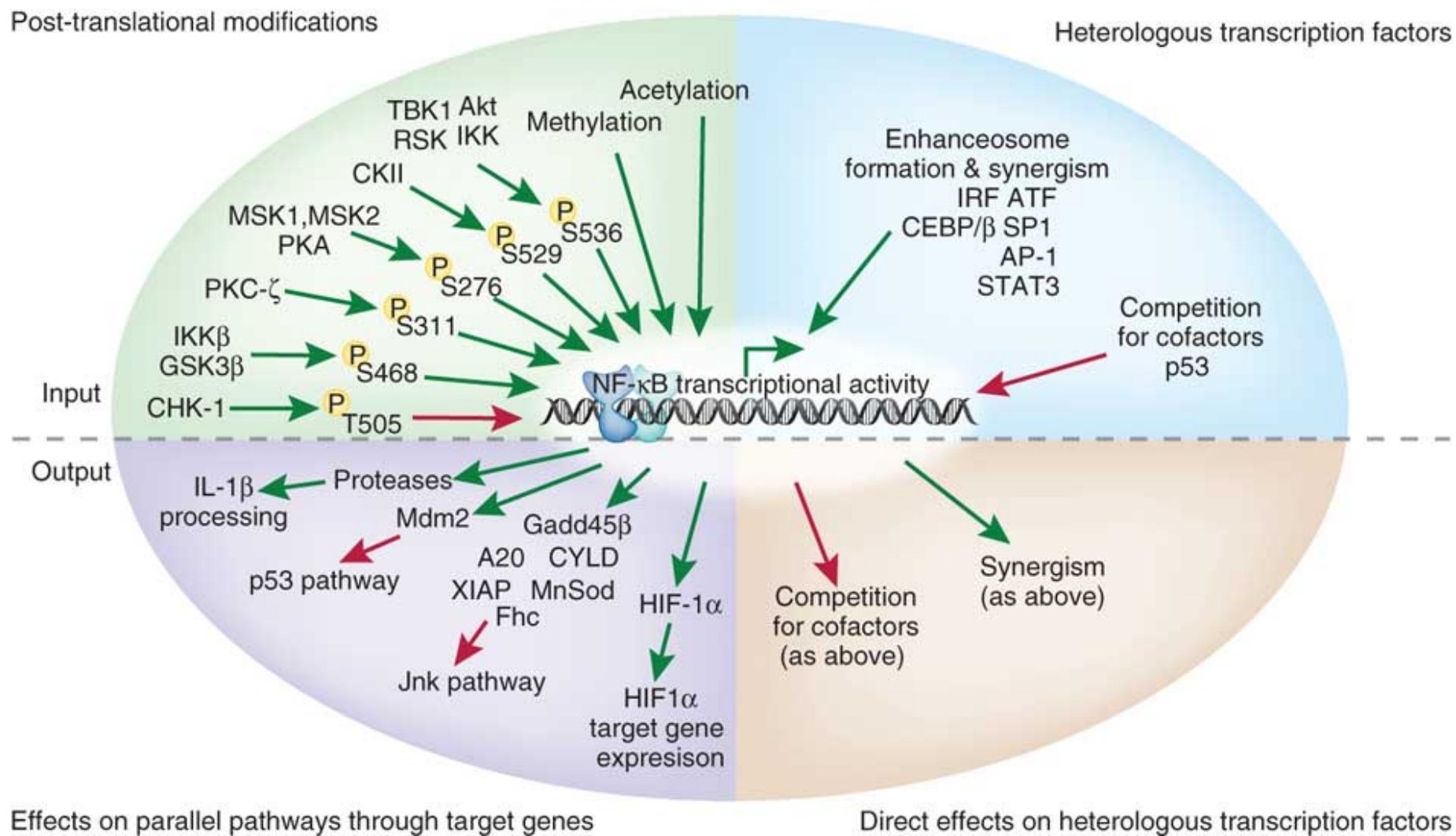
TRAF (TNF receptor associated factors), molecules playing key roles in the regulation (positive or negative) of the NF- κ B signaling pathway are also implicated in the regulation of other signaling pathways. They are hubs in the cross-talks of these different pathways.

Cross-talk of other signaling pathways with the inhibitor IKK



Several kinases can also phosphorylate IKKβ, demonstrating the large panel of stimuli that can activate NF-κB pathway through cross-talks. Moreover, IKKβ (and IKKα) can also phosphorylate other proteins that are not involved in the NF-κB pathway, thus leading to important consequences on biological functions.

Cross-talks linked to the transcriptional activity of NF- κ B



TD n°7 : Signaling pathways cross-talks

UE MMSC

Marc Davenne

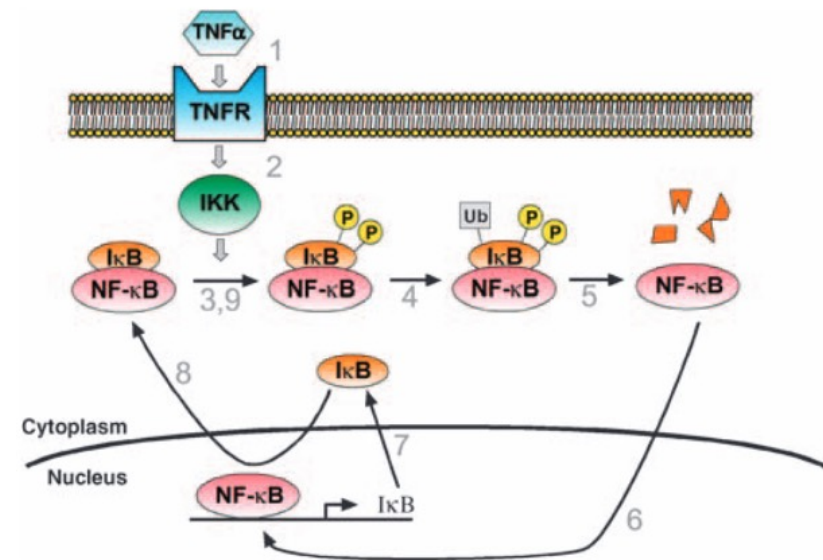
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Context of the study

Transcription factors of the NF- κ B family (p50 et p65) remain inactivated in the cytoplasm, associated with I κ B protein. Activation of this signaling pathway requires phosphorylation of I κ B by the IKK kinase complex, leading to I κ B ubiquitination, allowing its proteasome-mediated degradation. NF- κ B (p50 et p65) thus released from I κ B can translocate into the nucleus to activate its target genes.



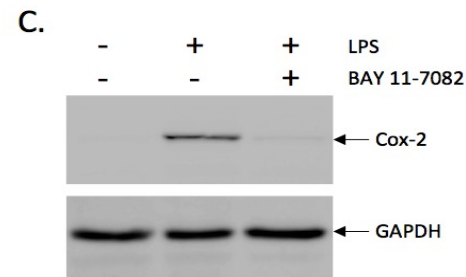
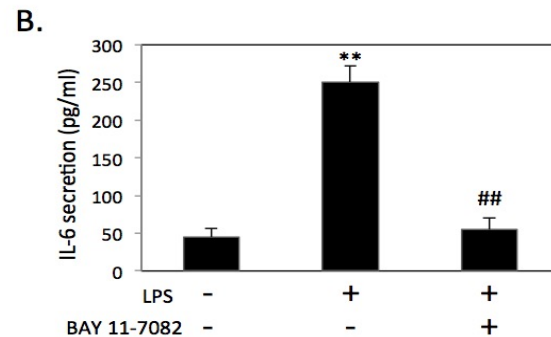
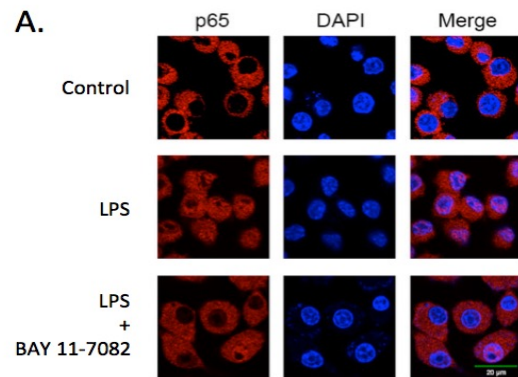
Cheong et al., 2008 doi:10.1038/msb.2008.30

Sujet d'examen de la deuxième session **20 janvier 2015**

In response to lipopolysaccharide (LPS) or Toll-Like receptors (TLR) agonists, RAW264 cells (macrophage cell line) produce a large panel of pro-inflammatory molecules such as IL-6 or cyclo-oxygenase 2 (COX-2).

The aim of this work is to study the NF- κ B signaling pathway in the inflammatory response of RAW264 cells and its regulation by cyclic AMP (cAMP).

Effect of IKK inhibition on LPS-induced IL-6 production and Cox-2 expression



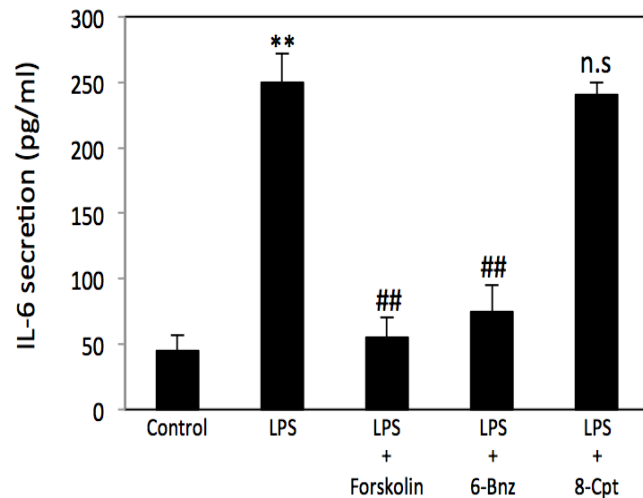
A. RAW264 cells were treated with LPS, with or without BAY 11-7082 (IKK inhibitor). Cells were fixed, permeabilized and immunostaining was performed with dsRed-coupled anti-p65 antibody. Nuclei were labeled with DAPI. **B.** IL-6 production was measured in culture media after 48 hours-treatment. **C.** Analysis of Cox-2 protein expression in RAW264 cells by Western blot. ** $P < 0.01$ compared to untreated condition and ## $P < 0.01$ compared to LPS treatment alone.

Question 1- Briefly describe the three experiments presented in Figure 1 (without going into technical details but specifying their relevance).

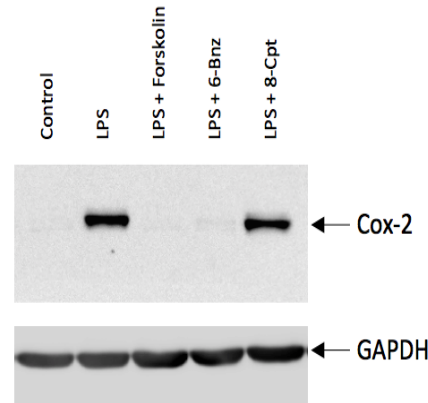
Question 2- Describe and analyze Figure 1 and conclude.

implication of cAMP in the NF- κ B-dependent inflammatory response of RAW264 cells induced by LPS

A.



B.

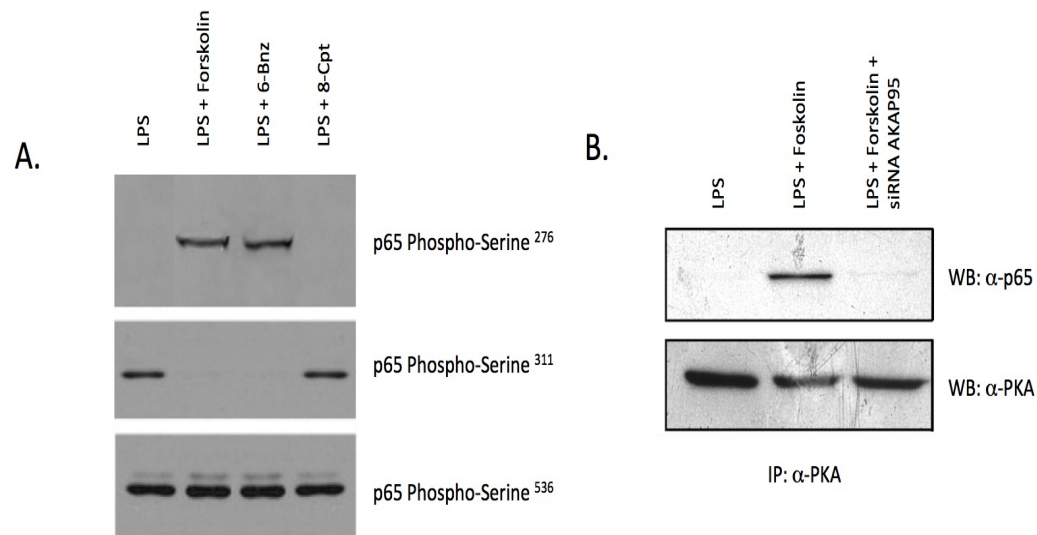


Cells were treated with LPS for 48h with or without forskoline (an adenylyl cyclase activator), 6-Bnz (a PKA activator) or 8-Cpt (an EPAC activator). **A.** IL-6 production was measured in culture media after 48 hours-treatment. **B.** Analysis of Cox-2 protein expression in RAW264 cells by Western blot. ** $P < 0.01$ compared to untreated condition and ## $P < 0.01$ compared to LPS treatment alone. n.s.=not significant.

Question 3- Based on your knowledge of the cAMP pathway, explain why authors used forskolin, 6-Bnz and 8-Cpt.

Question 4- Comment figure 2. Explain the possible role of cAMP on the LPS-induced inflammatory response of RAW264 cells.

cAMP regulation of LPS-induced inflammatory response of RAW cells



Cells were treated with LPS for 48h with or without forskoline, 6-Bnz or 8-Cpt.
A. P65 phosphorylation on S276, S311 and S536 assessed by immunoblotting.
B. PKA and p65 Co-immunoprecipitation. RAW264 cells were treated with LPS for 48h with or without forskolin and an siRNA targeting AKAP95. After cell lysis, an IP was performed using an anti-PKA antibody followed by an immunoblotting using anti-PKA and anti-p65 antibodies.

AKAP proteins (A-Kinase Anchor Protein) are anchor proteins for PKA.

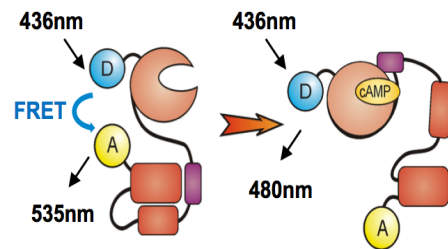
Question 5- Describe and analyze Figure **3A**. At this stage of the study, what hypothesis can you propose ?

Question 6. Which technique is used in Figure 3B? briefly describe it (eventually with a scheme). Analyze Figure 3B, what do you conclude?

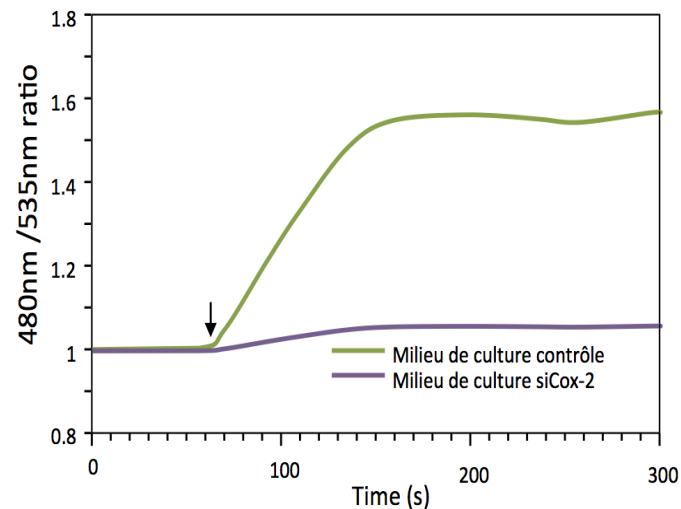
PGE2 implication in the cross-talk between LPS/NF- κ B and cAMP signaling pathways

Cox-2 expression was knocked-down by siRNA. RAW264 cells were then treated by LPS for 48 hours and culture media were then recovered (culture media from control cells treated with LPS: MC-Ctrl; and culture media from COX-2 siRNA-transfected cells, treated with LPS: MC-siCox-2). These culture media were then used on other RAW264 cells, expressing the $^{\text{T}}\text{EPAC}^{\text{VV}}$ biosensor.

A.



B.



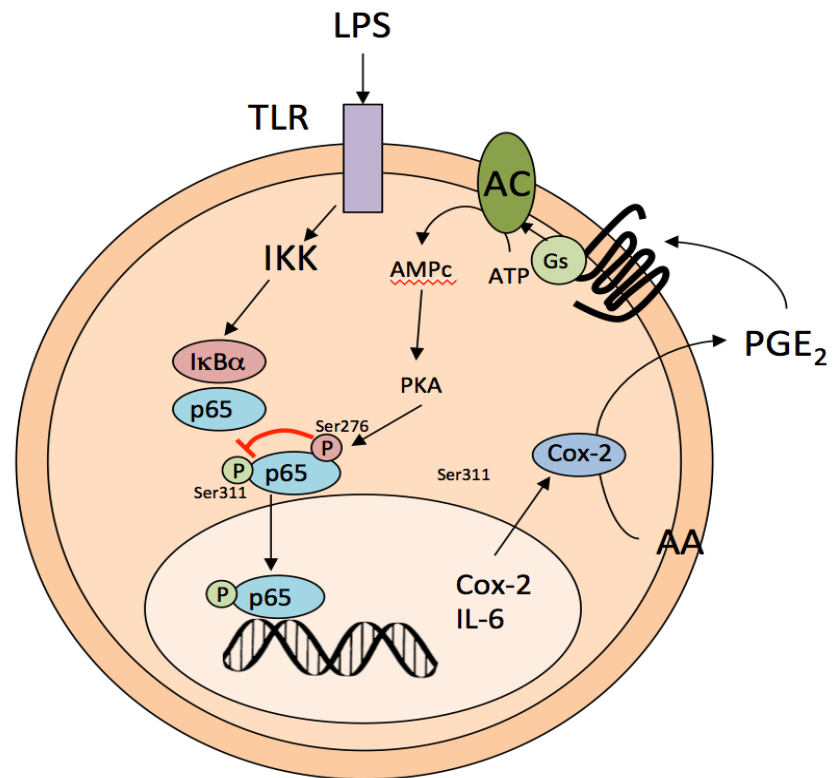
A. $^{\text{T}}\text{EPAC}^{\text{VV}}$ biosensor **B.** Dynamics of cAMP production in RAW264 cells. Naive RAW264 cells were perfused with basal media and then ($\downarrow$) with culture media from LPS-treated ctrl RAW264 cells (green curve) or LPS-treated RAW264 cells transfected with Cox-2 siRNA (purple curve).

Question 7 : Using the diagram in Fig4A, explain how a biosensor works.

Question 8: Comment and analyze figure 4B.

Question 9: Based on the results presented in this study, propose a summary scheme.

Summary



Effets anti-inflammatoires et immunosuppresseurs des glucocorticoïdes:
Cross-talks entre le récepteur de glucocorticoïdes (GR) et la voie NF κ B

Review

Therapeutic Mechanisms of Glucocorticoids

Jolien Vandewalle,^{1,2} Astrid Luybaert,^{3,4} Karolien De Bosscher,^{3,4} and Claude Libert^{1,2,*}

The Clinical Importance of GCs

GCs (Box 1) were discovered in the 1940s, a *tour de force* which was awarded the Nobel Prize in Physiology and Medicine in 1950. GCs are essential stress hormones that bind to the GC receptor, GR, and play a role in different fundamental processes such as metabolic homeostasis, cognition, mental health, cell proliferation, development, reproduction, and inflammation [1]. Owing to their anti-inflammatory and immune-suppressive actions, GCs are among the most widely prescribed drugs. Various autoimmune, inflammatory, and allergic disorders, such as rheumatoid arthritis (RA), lupus erythematosus, inflammatory bowel disease, transplant rejection and asthma, are often treated with synthetic GCs such as dexamethasone and prednisolone [1].

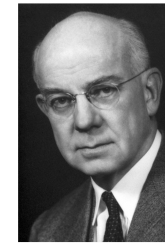


Photo from the Nobel Foundation archive.
Edward Calvin Kendall
Prize share: 1/3



Photo from the Nobel Foundation archive.
Tadeus Reichstein
Prize share: 1/3



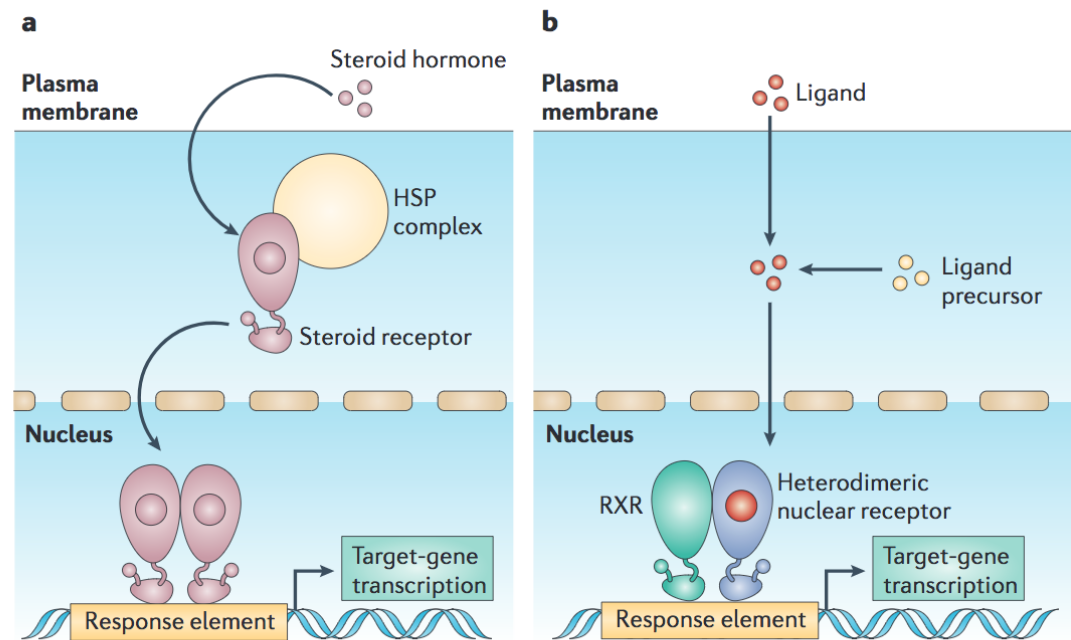
Photo from the Nobel Foundation archive.
Philip Showalter Hench
Prize share: 1/3



Rappel de cours : Les récepteurs nucléaires

Comme tous les RN des hormones stéroïdiennes, en l'absence de son ligand, GR est séquestré dans le cytoplasme par des protéines de choc thermique (HSP complex).

La liaison du ligand permet sa translocation au noyau où il peut se lier à ses éléments de réponse présents dans les promoteurs de ses gènes cibles. Il agit sous forme d'homodimères pour activer ou réprimer l'expression des gènes.



Glass, C., Ogawa, S. Combinatorial roles of nuclear receptors in inflammation and immunity. *Nat Rev Immunol* **6**, 44–55 (2006). <https://doi.org/10.1038/nri1748G>

Mécanismes d'action des effets anti-inflammatoires et immunosuppresseurs des glucocorticoïdes

Effets pharmacologiques	Mécanisme d'action	Conséquences biologiques
Effets anti-inflammatoires -Inhibition de la réparation des tissus -Antipyrétique -Antalgique -Inhibition de l'immunité innée	Inhibition de la production de cytokines pro-inflammatoires (IL-1, IL-6, IL-8, TNFalpha) Inhibition de l'expression de molécules d'adhésion (ICAM) Inhibition de la phospholipase A ₂ et de la cyclooxygénase de type 2 Inhibition de la NO synthase inducible	Diminution de l'afflux de macrophages et de granulocytes sur le site inflammatoire Diminution de la migration transendothéliale des cellules phagocytaires Inhibition de la synthèse d'eicosanoïdes pro-inflammatoires (Prostaglandines, thromboxane, leucotriènes) Diminution de la production d'espèces radicalaires
Effets immunosuppresseurs (Immunité acquise)	Diminution de l'expression des molécules du CMH II par les macrophages Inhibition de la production d'IL-2 par les lymphocytes T	Diminution de l'antigénicité des protéines Diminution de la prolifération lymphocytaire

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	Inhibition de l'expression de molécules d'adhésion (ICAM)	Diminution de la migration transendothéliale des cellules phagocytaires
	Inhibition de la phospholipase A ₂ et de la cyclooxygénase de type 2	Inhibition de la synthèse d'eicosanoïdes pro-inflammatoires (Prostaglandines, thromboxane, leucotriènes)
	Inhibition de la NO synthase inducible	Diminution de la production d'espèces radicalaires
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L'article qui a initié cette démonstration fait l'objet de ce TD

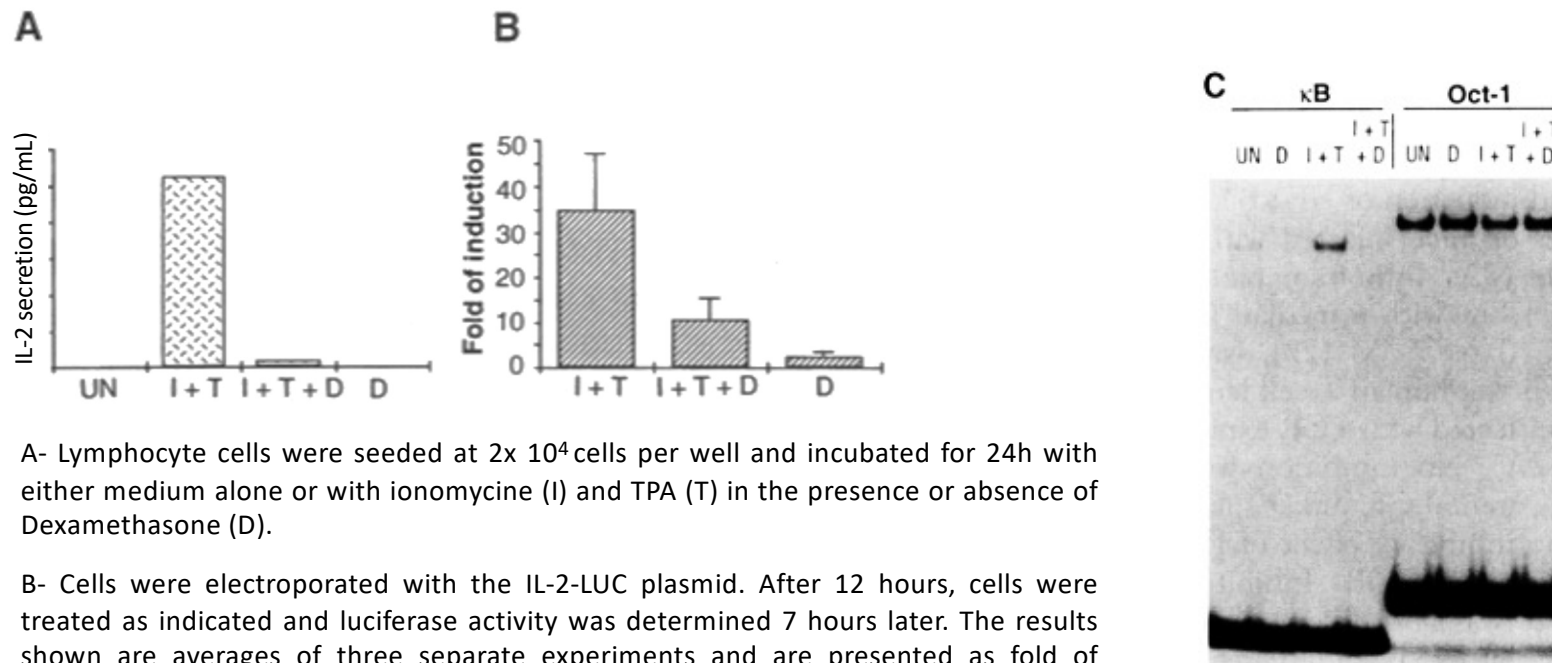
Contexte de l'étude

Hypothèse : les glucocorticoïdes pourraient inhiber l'expression du gène codant pour l'IL2 en interférant avec l'activité du facteur de transcription pro-inflammatoire NF- κ B.

Ce qui est connu:

- L'expression du gène IL2 est sous le contrôle des facteurs de transcription NF- κ B, Oct-1 et AP1.
- NF- κ B est séquestré dans le cytoplasme par I κ B.

Role of GR in IL2 expression and secretion



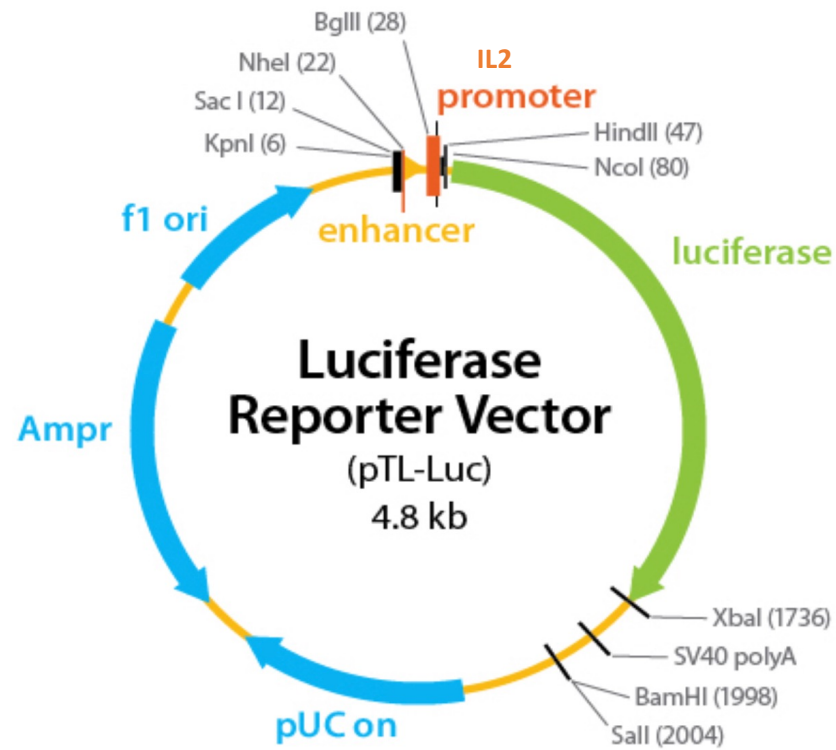
A- Lymphocyte cells were seeded at 2×10^4 cells per well and incubated for 24h with either medium alone or with ionomycin (I) and TPA (T) in the presence or absence of Dexamethasone (D).

B- Cells were electroporated with the IL-2-LUC plasmid. After 12 hours, cells were treated as indicated and luciferase activity was determined 7 hours later. The results shown are averages of three separate experiments and are presented as fold of induction above basal IL-2-LUC expression.

C- Cells were treated as indicated and after 1 hour, nuclear extracts were prepared. Five micrograms of each extract were incubated with 20.000 cpm of ^{32}P -labeled NF-κB, Oct-1 binding site probes, and DNA binding activity was measured by EMSAs.

Analysez chacune des figures (objectif, matériel et méthodes, résultats, conclusion)

Structure of a Luciferase reporter vector

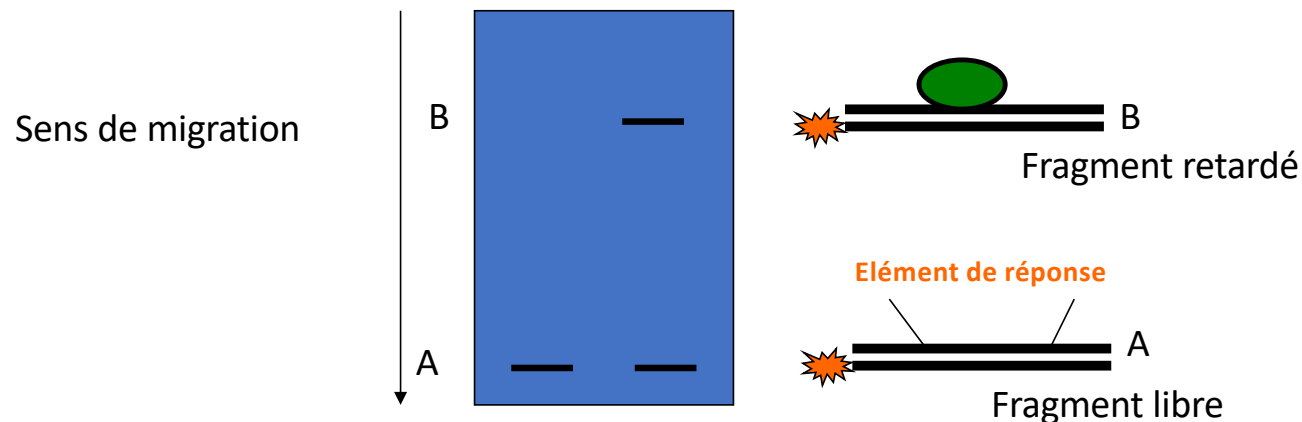


Le retard sur gel (Gel shift ou EMSA: *electrophoretic mobility shift assay*)

Objectif: permet de mettre en évidence des protéines ou des complexes protéiques se liant à une séquence d'ADN donnée (située dans la région promotrice)

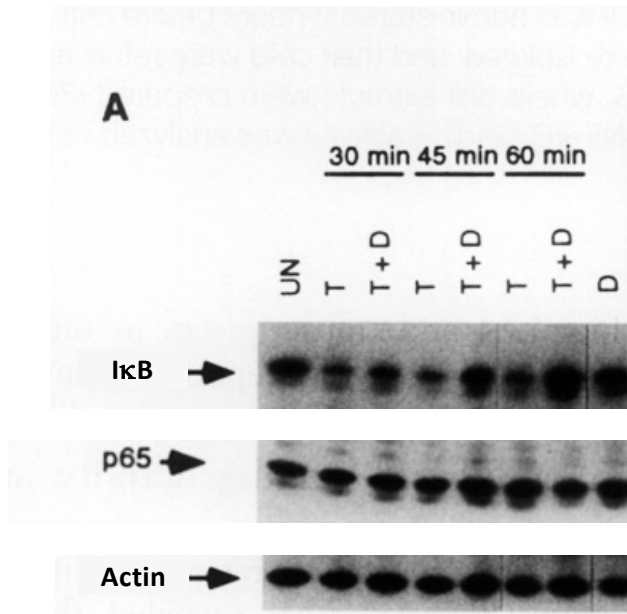
Etapes:

- Extraction des protéines à partir des noyaux du tissu d'intérêt (extraits nucléaires)
 - Incubation avec le fragment d'ADN radioactif portant la région du promoteur d'intérêt
 - Séparation des fragments par migration sur un gel de polyacrylamide non dénaturant:
- Les complexes ADN-protéines sont retardés par rapport à la sonde libre

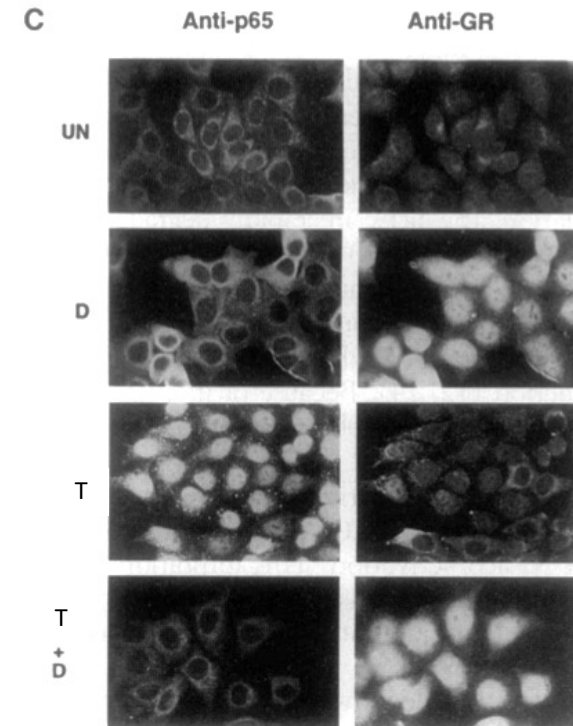


Autoradiogramme d'un retard sur gel

Role of GR in NF- κ B regulation



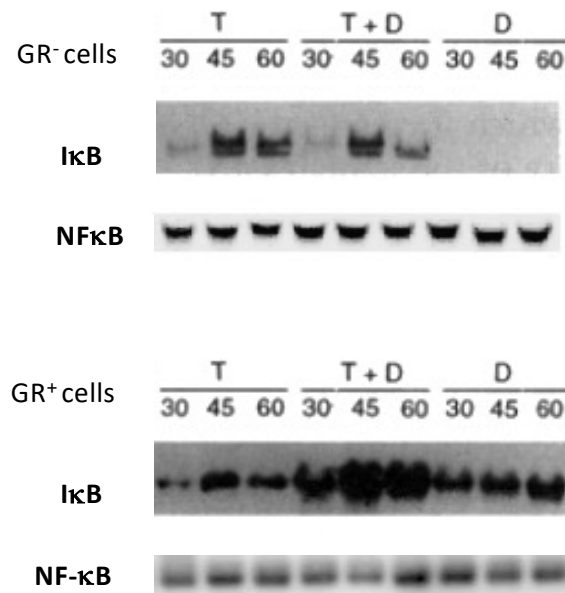
A- After different treatment durations, cells were lysed and 40 μ g of each lysate was separated by SDS-polyacrylamide gel electrophoresis (PAGE), transferred to a polyvinyl membrane, and immunoblotted with either anti-I κ B, anti-p65 or anti-actin. UN: Untreated, T: TPA + ionomycin, D: Dexamethasone.



C- Cells were plated on glass coverslips and were untreated (UN) or were treated with DEX (D), TPA + ionomycin (T) or with TPA + ionomycin (T) and DEX (D). After treatment, the cells were fixed and revealed by indirect immunofluorescence with the use of anti-p65, and anti-GR.

Analysez chacune des figures (objectif, matériel et méthodes, résultats, conclusion)

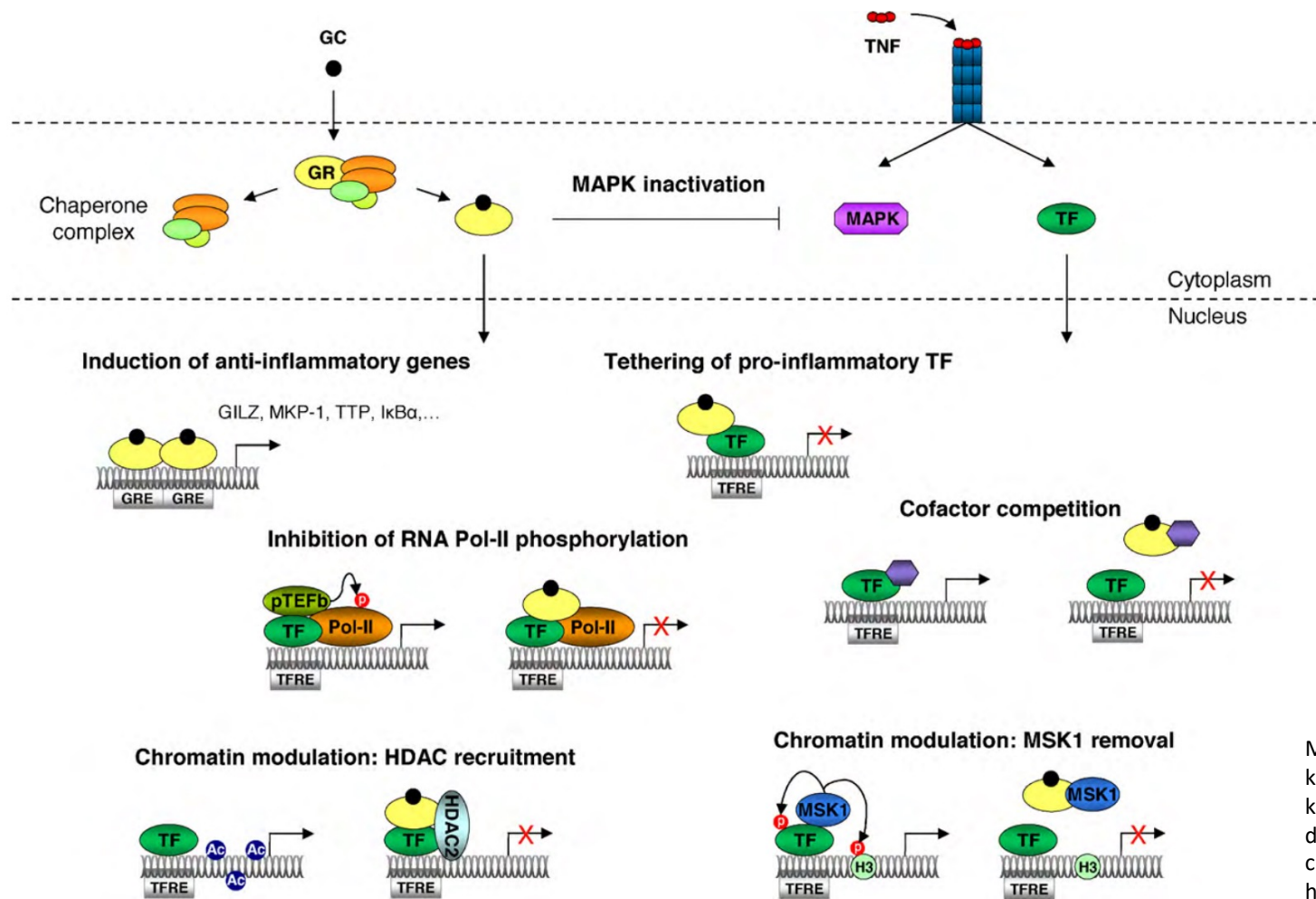
Role of GR in I κ B expression



GR⁻ and GR⁺ cells were treated for different lengths of time (in minutes), as indicated. Total cytoplasmic RNA was extracted and 20 μ g samples were separated on a 1% agarose gel, transferred to a nylon membrane and analyzed by hybridization (I κ B and p65) complementary DNA probes.

Analysez la figure (objectif, matériel et méthodes, résultats, conclusion)

Les autres mécanismes d'interférence de GR avec des FT pro-inflammatoires



Mitogen- and stress-activated kinase 1 (MSK1) is a chromatin kinase that facilitates activator-dependent transcription by altering chromatin structure through histone H3 phosphorylation.